

REMARCS: A NEW APPROACH FOR ANALYZING HIGH-FREQUENCY REPEATED
CROSS-SECTIONAL DATA FROM HOUSEHOLD FOOD SECURITY SURVEYS

AN ABSTRACT

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Abstract

Repeated cross-sectional (RCS) surveys are widely employed by government agencies and organizations to monitor and model evolving metrics. When conducted at high frequencies, these surveys are prone to be temporally correlated. Existing methods for modeling data of this type either rely on time-step aggregates that result in information loss or fail to properly account for variability when estimating model parameters. In response to this gap in existing methods, we have developed REMARCS (REgression Models for Autocorrelated Repeated Cross-Sections). The accompanying method employs an iterated GLS estimator that fits residuals with an ARMA-Noise model generalized to accommodate variable sample sizes. The efficacy of this algorithm is demonstrated both with simulated data and a case study involving household food security survey data collected by the World Food Programme.

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As much time as I may have spent on the classes and research that have informed this document, the past five years will forever be defined by the people with whom I spent them. When I first arrived in New Orleans, I had signed a lease with a couple of strangers, Joseph and Lily, who in no time would turn into dear friends. Eager to give my all to this PhD program, I decided I would ensure I was working hard by getting to the math department as early as six in the morning. While this was a practice that lasted little more than a semester, it did lead to becoming friends with the other early birds of the building. I'd like to especially acknowledge Ms. Tania Gomez, who became my unofficial Spanish teacher and official department mom, displaying endless patience and support that allowed me to nurture my love for language learning at a time when there wasn't room for a more formal pursuit. When I think of the places that live most vividly in my mind, few in the city hold the place that the restaurant Singleton's does. For me, the value of a hole-in-the-wall serving delicious comfort food and even better company cannot be understated. To Bau and Laura, I thank you from the bottom of my heart for all you've done for me and this city.

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stakes exams, it was the snack-filled study sessions in empty classrooms that taught me more than most lectures. I'd like to especially acknowledge Ketan Kalgi for being my study buddy from day one and Lan Trinh for being a constant inspiration.

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I consider myself immeasurably lucky to have lifelong friends like Chris Petruzzo and Jonathan Rub, friends who have seen me at my best and worst and given the gift of their friendship through it all. To all those I haven't named, I'm looking forward to conveying my gratitude through all the experiences we share from here on out. However, there is nobody who played a bigger role in this part of my life than

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The past five years have changed me in so many way and have made New Orleans a place I call home. It is this place that I'd like to thank lastly, this place that is filled with so much magic and that rewards so richly those that give themselves over to it. Whatever comes next, I'll know what it means to love New Orleans.

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Chapter 1

Introduction

Mobile phone surveys are an increasingly popular method for tracking household food security that is employed by aid organizations like the World Food Programme. Improving sampling capabilities and enabling real-time tracking of consumption and coping metrics, these surveys are conducted daily in an array of countries. Drawn with a random sampling strategy to select participants, samples do not necessarily overlap day to day, but are conducted at a frequency that often results in temporal correlation in the collected responses that hinders inference when using linear regression. While there already exist methods for analyzing data in which samples are surveyed repeatedly or aggregated at the time-step level, models that account for individual-level effects and population-level shocks are scarce. Those that do exist model temporal fluctuations with a latent time series that is recursively estimated and used to update parameter estimates via an EM algorithm, but do not properly account for how cross-sectional noise impacts the analysis thereof. We correct this shortcoming by incorporating methods for analyzing data that are the result of multiple noisy observations of each element of a time series, known as an ARMA-Noise model.

We go on to establish that the use of an EM algorithm that factors the likeli-

hood equation by conditioning on an estimate of a latent ARMA process incorrectly assumes that the resulting deviations will have a covariance structure conducive to standard fixed effect regression. We therefore replace the EM algorithm with one that recursively estimates variance parameters with an ARMA-Noise model and uses them to calculate GLS estimators for regression parameters. Confidence regions for time series parameters are generated via the Delta Method due to their latent nature. The respective implications of varying orders of the latent time ARMA process inform a decision rule for model selection.

While the newly incorporated method for fitting time series parameters corrects the bias that previously existed, it imposes a constant sample size at the time-step level, an unrealistic requirement in real-life surveying contexts. We therefore explored the extent to which ARMA-Noise modeling methods could be generalized to accommodate varying sample sizes.

Given the size of the datasets motivating this work, computational methods were developed to expedite the process. These methods leverage the relationship between the spectral decompositions of the covariance matrices of the observed data and latent time series.

With the proposed method fully developed, we test on simulated data and illustrate how model selection might be performed on a realization from a restricted model. We also include a case study on household survey data from Honduras that shows the method’s potential to decorrelate residuals. The results of our analysis support the method as a viable option for modeling RCS survey data related to food security and other contexts.

Chapter 2

Time Series

2.1 Time Series Fundamentals

We begin by introducing some machinery and notation from time series analysis before detailing existing methods. Unless otherwise stated, all results in this chapter are from Brockwell and Davis (1991) [3]. If $\{X_t\}_{t=1}^T$ is a time series such that $\text{Var}(X_t) < \infty \forall t$, then the autocovariance function (ACVF) $\gamma_X(\cdot, \cdot)$ is defined by $\gamma_X(r, s) = \text{Cov}(X_r, X_s)$, $r, s \in \{t\}$

A time series $\{X_t\}_{t \in \mathbb{Z}}$ is *weakly stationary* if:

- i $E|X_t|^2 < \infty$ for all t
- ii $EX_t = \mu$ for all t
- iii $\gamma_X(r, s) = \gamma_X(r + h, s + h)$ for all r, s, t

With these conditions satisfied, the ACVF can be more simply parameterized as $\gamma_X(h) := \text{Cov}(X_t, X_{t+h})$. The autocorrelation function (ACF) $\rho_X(h)$ of a stationary time series is then defined as $\rho_X(h) = \gamma_X(h)/\gamma_X(0)$.

A time series is *strictly stationary* if:

$$(X_{t_1}, \dots, X_{t_k})' \stackrel{d}{=} (X_{t_1+h}, \dots, X_{t_k+h}) \text{ for all } k \in \mathbb{N} \text{ and } t_1, \dots, t_k, h \in \mathbb{Z}$$

For the purposes of this dissertation, “stationary” will refer to the weak type. Furthermore, a time series is Gaussian if all finite subsets of $\{X_t\}$ have a multivariate normal distribution. Notably, a Gaussian time series is strictly stationary if it is weakly stationary.

A time series is said to be white noise with mean 0 and variance σ^2 if and only if:

$$\gamma_X(h) = \begin{cases} \sigma^2, & h = 0 \\ 0, & h \neq 0 \end{cases} \quad (2.1.1)$$

The ACVF is estimated with the Sample Auto-Covariance Function (ACVF), defined by:

$$\hat{\gamma}_X(h) := n^{-1} \sum_{j=1}^{n-h} (x_{j+h} - \bar{x})(x_j - \bar{x}), \quad 0 \leq h \leq n-1 \quad (2.1.2)$$

The choice of dividing by n and not $(n-h)$ in Equation 2.1.2 ensures that the $n \times n$ matrix $\hat{\Gamma}$, the i, j entry of which is $\hat{\gamma}_X(i-j)$, is always nonsingular. The Sample Auto-Correlation Function (SACF) is then defined as $\hat{\rho}_X(h) = \hat{\gamma}_X(h)/\hat{\gamma}_X(0)$

2.2 ARMA Processes

A time series $\{Y_t\}_{t \in \mathbb{Z}}$ is a linear process if, for all t , $Y_t = \sum_{j=-\infty}^{\infty} \psi_j Y_{t-j}$ and $\sum_{j=-\infty}^{\infty} |\psi_{t-j}|$ converges. We can express Y_t with $\sum_{j=-\infty}^{\infty} \psi_j B^j Y_t$, introducing the backwards operator B , which behaves as follows: $(B^a)Y_t = Y_{t-a}$. A stationary time series $\{Y_t\}$ is an auto-regressive moving average process of order p, q (denoted $\text{ARMA}(p, q)$) if it is expressible as:

$$\phi(B)Y_t = \theta(B)Z_t, \quad Z_t \sim WN(0, \sigma^2), \quad (2.2.1)$$

in which $\phi(B)$ and $\theta(B)$ are polynomials in B of orders p and q , respectively. Importantly, the two polynomials must share no common roots and have no roots of magnitude 1. The process is *causal* if all roots of $\phi(B)$ are of magnitude strictly less than 1, as this will result in $\psi_j = 0$ for all $j < 0$, meaning Y_t is only a function of “past” values. The time series is *reversible* if all roots of $\theta(B)$ are of magnitude strictly less than 1. We will be restricting the scope of this dissertation to causal, reversible ARMA processes for which $p, q \leq 1$. It will therefore always be the case that $|\phi|, |\theta| < 1$.

If $\{Y_t\}$ is an AR(1) process, it is defined by the equation $Y_t - \phi Y_{t-1} = Z_t$ with $Z_t \stackrel{\text{i.i.d.}}{\sim} WN(0, \sigma^2)$ and has the following auto-covariance function:

$$\gamma_Y(h) = \frac{\sigma^2 \phi^{|h|}}{1 - \phi^2}, \quad h \in \mathbb{Z} \quad (2.2.2)$$

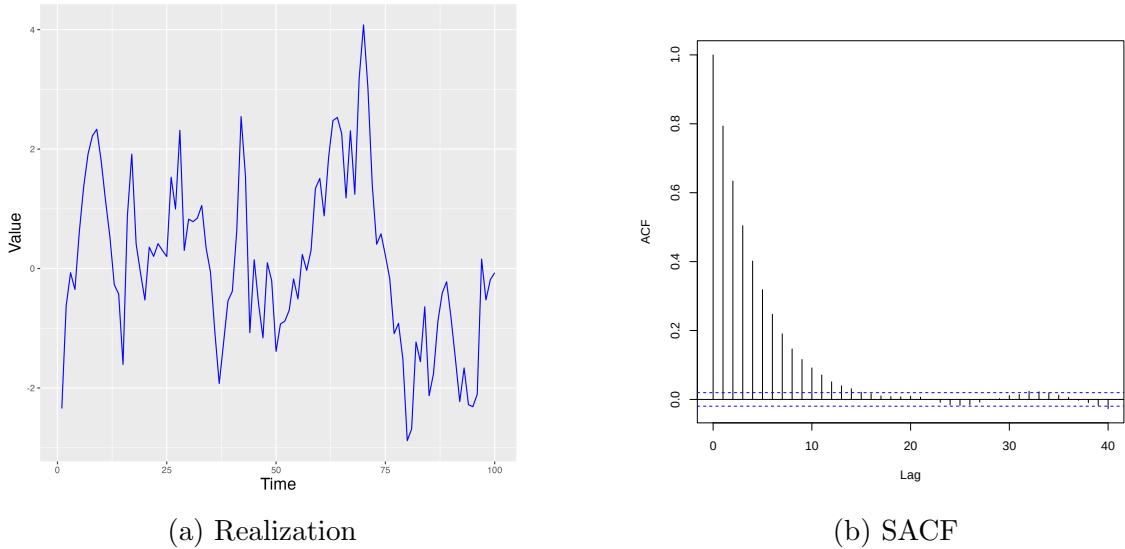


Figure 2.1: Sample AR(1) Process with $(\phi, \sigma^2) = (0.8, 1)$

In Figure 2.1, we see a sample AR(1) process with a strong positive coefficient.

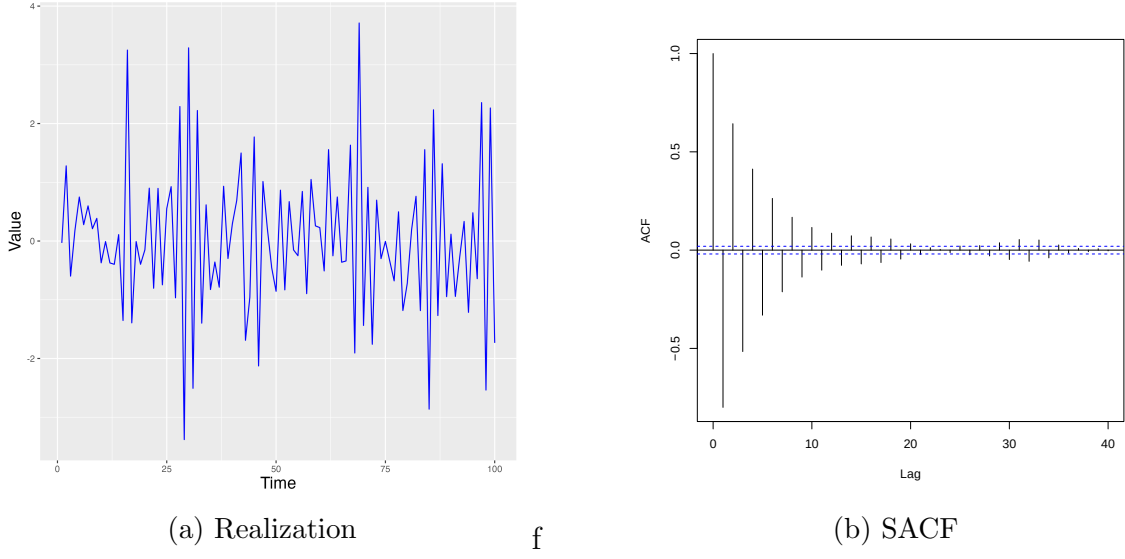


Figure 2.2: Sample AR(1) Process with $(\phi, \sigma^2) = (-0.8, 1)$

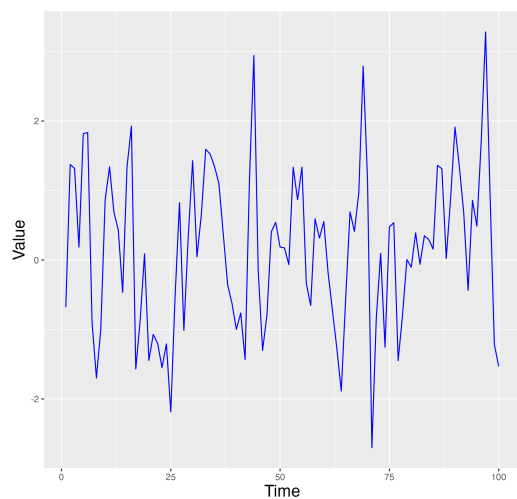
By contrast, an AR(1) with a strong negative coefficient will oscillate rapidly, as shown in Figure 2.2.

An MA(1) process, which can be expressed as an equation of the form $Y_t = Z_t + \theta Z_t$, $\{Z_t\} \sim \text{WN}(0, \sigma^2)$ has the following ACVF:

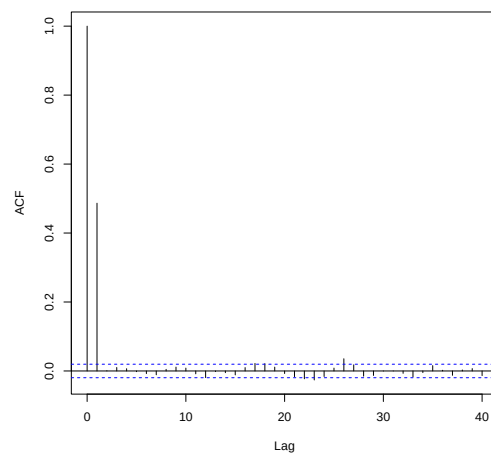
$$\gamma_Y(h) = \begin{cases} (1 + \theta^2)\sigma^2 & h = 0 \\ \theta\sigma^2 & |h| = 1 \\ 0 & \text{else} \end{cases} \quad (2.2.3)$$

Figures 2.3 and 2.4 illustrate MA(1) processes with $\sigma^2 = 1$ and $\theta = 0.8$ and $\theta = -0.8$, respectively.

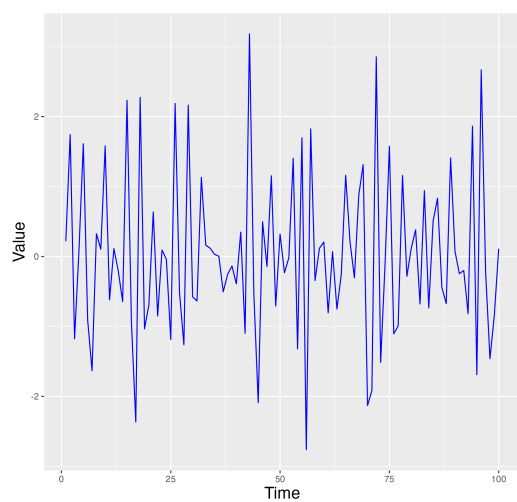
An ARMA(1,1) process with the form $Y_t - \phi Y_t = Z_t + \theta Z_{t-1}$ has the following ACVF:



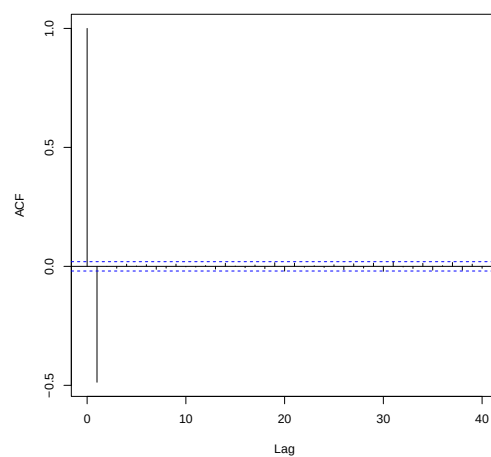
(a) Realization



(b) SACF

Figure 2.3: Sample MA(1) Process with $(\theta, \sigma^2) = (0.8, 1)$ 

(a) Realization



(b) SACF

Figure 2.4: Sample MA(1) Process with $(\theta, \sigma^2) = (-0.8, 1)$

$$\gamma_Y(h) = \begin{cases} \sigma^2(1 + \frac{(\phi+\theta)^2}{1-\phi^2}) & h = 0 \\ \sigma^2(\phi + \theta + \frac{\phi(\phi+\theta)^2}{1-\phi^2}) & |h| = 1 \\ \phi^{|h|-1}\gamma_Y(1) & |h| > 1 \end{cases} \quad (2.2.4)$$

Recall that we can express the equation governing an ARMA(1,1) process $\{Y_t\}$ as follows:

$$(1 - \phi B)Y_t = (1 + \theta B)Z_t$$

Because we are imposing the restriction that $|\phi| < 1$, we can generate an explicit equation for Y_t :

$$Y_t = (1 - \phi B)^{-1}(1 + \theta B)Z_t, \quad (1 - \phi B)^{-1}X_t = \sum_{j=0}^{\infty} (\phi B)^j X_t = \sum_{j=0}^{\infty} \phi^j X_{t-j}$$

2.3 Parameter Estimation for ARMA processes

When fitting parameter estimates to a realization of an ARMA(1,1) model, we can leverage a transformed version of the likelihood equation based on the one step predictors $\hat{X}_{t+1} = \mathbb{E}(X_{t+1}|X_t, X_{t-1}, \dots, X_1)$.

Let Σ_X be the $T \times T$ covariance matrix of the zero-mean ARMA(1,1) Gaussian process $\{X_t\}$. The likelihood equation is then given by:

$$L(\phi, \theta, \sigma^2|\mathbf{X}) = (2\pi)^{-T/2} |\Sigma_X| \exp \{ \mathbf{X}' \Sigma_X^{-1} \mathbf{X} \}$$

From Brockwell and Davis [3], we can rewrite this likelihood as follows:

$$L(\phi, \theta, \sigma^2 | \mathbf{X}) = (2\pi\sigma^2)^{-T/2} (r_0 - \dots r_{T-1})^{-1/2} \exp \left\{ -\frac{1}{2\sigma^2} \sum_{j=1}^T \frac{(X_j - \hat{X}_j)^2}{r_{j-1}} \right\}$$

The one step predictors $\{\hat{X}_t\}_{t=1}^T$ and their corresponding mean squared errors $\{r_t = \sigma^{-2} \mathbb{E}(X_{t+1} - \hat{X}_{t+1})^2\}_{t=0}^{T-1}$ are given by the Innovations Algorithm:

$$\begin{cases} \hat{X}_1 = 0 \\ r_0 = \sigma^{-2} \gamma_X(0) = 1 + \frac{(\phi+\theta)^2}{(1-\phi^2)} \\ \theta_{t1} = \theta/r_{n-1} \\ \hat{X}_{t+1} = \phi X_t + \theta_{n1}(X_t - \hat{X}_t) \\ r_t = 1 + \theta^2 - \theta^2/r_{t-1}, \quad t \geq 1 \end{cases} \quad (2.3.1)$$

This recursive sequence of coefficients can be initially inaccessible, but we see in Figure 2.5 that θ_{t1} and r_t both rapidly approach θ and 1, and their respective values for small values of t account for the lack of history at the start of the time series.

It doesn't take long for the one step predictors detailed in Equation 2.3.1 to be effectively equivalent to $\hat{X}_{t+1} = \phi X_t + \theta(X_t - \hat{X}_t)$ with $\mathbb{E}(X_t - \hat{X}_t)^2 = \sigma^2$. We now derive the MLE for σ^2 using this new expression of the likelihood:

$$l(\phi, \theta, \sigma^2) = -\frac{T}{2} \ln(2\pi\sigma^2) - \frac{1}{2} \ln \left(\prod_{t=1}^T r_{t-1} \right) - \frac{1}{2\sigma^2} \sum_{t=1}^T \frac{(X_t - \hat{X}_t)^2}{r_{t-1}} \quad (2.3.2)$$

$$l(\phi, \theta, \sigma^2) = -\frac{T}{2} \ln(2\pi\sigma^2) - \frac{1}{2} \sum_{t=1}^T \ln(r_{t-1}) - \frac{1}{2\sigma^2} \sum_{t=1}^T \frac{(X_t - \hat{X}_t)^2}{r_{t-1}}$$

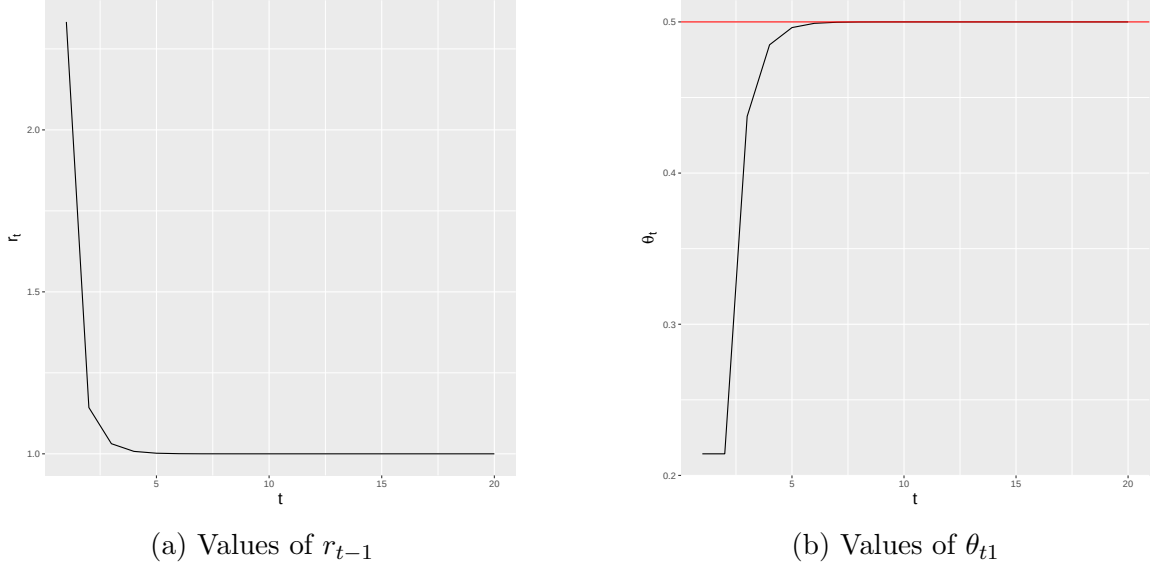


Figure 2.5: Coefficients provided by Innovations Algorithm for Calculating One-Step Predictors of ARMA(1,1) process

Noting that $\{r_t\}$, $\{\theta_{t1}\}$ and consequently $\{\hat{X}_t\}$ are all functions independent of σ^2 , we obtain the following derivative:

$$\frac{\partial}{\partial \sigma^2} l(\phi, \theta, \sigma^2) = -\frac{T}{2\sigma^2} + \frac{1}{2\sigma^4} \sum_{t=1}^T \frac{(X_t - \hat{X}_t)^2}{r_{t-1}} \equiv 0$$

Defining $S(\phi, \theta) := \frac{\sum_{t=1}^T (X_t - \hat{X}_t)^2}{r_{t-1}}$, we know $\hat{\sigma}_{\text{ML}}^2$ must satisfy:

$$\hat{\sigma}_{\text{ML}}^2 = \frac{S(\hat{\phi}, \hat{\theta})}{T}$$

Substituting this back into our log likelihood, we obtain:

$$l(\phi, \theta, \sigma^2) = -\frac{T}{2} \ln(2\pi) - \frac{T}{2} \ln(T^{-1} S(\phi, \theta)) - \frac{1}{2} \sum_{t=1}^T \ln(r_{t-1}) - \frac{T}{2S(\phi, \theta)} S(\phi, \theta)$$

$$l(\phi, \theta, \sigma^2) = -\frac{T}{2} \left[\ln(2\pi) + \ln(T^{-1}S(\phi, \theta)) + \frac{1}{2} \sum_{t=1}^T \ln(r_{t-1}) + \frac{1}{S(\phi, \theta)} S(\phi, \theta) \right]$$

By removing the constant terms we can deduce that $\hat{\phi}$ and $\hat{\theta}$ must minimize the expression:

$$\ln(T^{-1}S(\phi, \theta)) + \frac{1}{2} \sum_{t=1}^T \ln(r_{t-1})$$

Doing so is a numerical process that is greatly facilitated by good initial estimates, which can be calculated with such methods as the Durbin-Levinson Algorithm, the Innovations Algorithm, and/or the Yule-Walker Algorithm depending on the order of the time series [3]. The asymptotic behavior of the Maximum Likelihood Estimator and its implications on inference are discussed in the next section.

2.4 Inference for ARMA Processes

For an MA(1) process parameterized by θ and σ_η^2 , the normalized MLE has the following asymptotic distribution:

$$\sqrt{T}(\hat{\theta} - \theta) \xrightarrow{d} N(0, \sigma_\theta^2), \quad \sigma_\theta^2 = 1 - \theta^2 \quad (2.4.1)$$

For an AR(1) process parameterized by ϕ and σ_η^2 , the asymptotics are similar:

$$\sqrt{T}(\hat{\phi} - \phi) \xrightarrow{d} N(0, \sigma_\phi^2), \quad \sigma_\phi^2 = 1 - \phi^2 \quad (2.4.2)$$

For an ARMA(1,1) process parameterized by ϕ , θ , and σ_η^2 , we see asymptotic covariance in the normalized estimate:

$$\sqrt{T} \left[\begin{pmatrix} \hat{\phi} \\ \hat{\theta} \end{pmatrix} - \begin{pmatrix} \phi \\ \theta \end{pmatrix} \right] \xrightarrow{d} N(\mathbf{0}, \Sigma_{\phi, \theta})$$

$$\Sigma_{\phi, \theta} = \frac{1 + \phi\theta_b}{(\phi + \theta_b)^2} \begin{bmatrix} (1 - \phi^2)(1 + \phi\theta_b) & -(1 - \theta_b^2)(1 - \phi^2) \\ -(1 - \theta_b^2)(1 - \phi^2) & (1 - \theta_b^2)(1 + \phi\theta_b) \end{bmatrix} \quad (2.4.3)$$

In each of these cases, the normalized estimate for the variance parameter σ_η^2 is asymptotically independent of any AR or MA parameters. While this may not be immediately obvious, recall that the maximum likelihood estimators for ϕ and θ are based on $S(\phi, \theta)$, a statistic independent of σ_η^2 . The estimator $\hat{\sigma}_\eta^2 = T^{-1}S(\hat{\phi}, \hat{\theta})$ has the following asymptotic distribution:

$$\sqrt{T}(\hat{\sigma}_\eta^2 - \sigma_\eta^2) \xrightarrow{d} N(0, 2\sigma_\eta^4)$$

These asymptotic covariance matrices enable the construction of confidence regions centered at parameter estimates. With this machinery in place, we next review some of the methods that already exist for analyzing repeated cross-sectional data.

Chapter 3

Repeated Cross Sectional Data

Having detailed some useful machinery for analyzing univariate time series, we now consider data obtained through multiple observations, each with i.i.d noise, made at each time step. Such data are defined as repeated cross-sections (RCS), and they have long been the subject of analysis. When cross-sections are taken with high frequency, the possibility of temporal correlation arises. While panel data consist of a single sample being observed repeatedly, RCS data do not necessarily have any repeated observations, ruling out the treatment of the data as a multivariate time series. It is often instead the population mean that is treated as a time series, with individual observations modeled as a realization of that mean with some added noise.

3.1 Modeling RCS Data with a Univariate Time Series

Given such a dataset, it may seem apt to simply calculate averages at each time step and analyze these averages as a time series with covariates. Such a model is an example of a state-space model that can be modeled jointly via Kalman filtering [3]. In the regression setting, aggregates would be needed for both response and

explanatory variables, fitting the following model:

$$\bar{\mathbf{y}} = (n^{-1}X_t'X_f)\beta + \mathbf{u}, \quad \mathbf{u} \sim \text{ARMA}(p,q)$$

This method is very effective at teasing out population-level fixed effects like intercepts and slopes, but is limited when it comes to the estimation of individual-level effects. A scenario in which this method falls short is the presence of indicator variables that vary little in their distribution at each time step. This is a highly plausible scenario when it comes to sampling methods for surveys. If researchers are interested in the effect of a certain binary covariate, for example, they may very well aim to be consistent in the proportions of time step samples that have this feature. Such consistency hinders the ability of the significant effect to be estimated as such, which we illustrate through a simulated simple regression model with a trend and autocorrelated errors:

$$y_{ti} = \beta_0 + \beta_1 t + \beta_2 x_{ti} + u_t + \varepsilon_{ti}, \quad i = 1, \dots, n_t$$

$$u_t - \phi u_{t-1} = \eta_t + \theta \eta_{t-1}, \quad t = 1, \dots, T$$

$$\eta_t \stackrel{\text{i.i.d.}}{\sim} N(0, \sigma_\eta^2), \quad x_{ti} \sim \text{Bernoulli}(.5), \quad \varepsilon_{ti} \stackrel{\text{i.i.d.}}{\sim} N(0, \sigma_\varepsilon^2)$$

We consider fitting this model by aggregating the data at each time point and treating the averages as a time series with covariates. 100 simulations were conducted with the following parameters:

$$\beta = (43, -.001, .25), \quad \sigma_\varepsilon^2 = 100, \quad \phi = \theta = .5, \quad \sigma_\eta^2 = 1$$

Some sample rows of simulated data are provided below:

t	x	y
1.00	0	48.14
1.00	0	25.17
1.00	0	42.60
1.00	1	68.78
1.00	0	37.36
1.00	0	30.96

After aggregating at each time step, we obtain the following condensed dataset, from which a time series with covariates is then fit:

t	n	x	$\bar{y}$
1.00	94	0.37	44.88
2.00	84	0.50	43.24
3.00	96	0.47	43.49
4.00	107	0.52	42.68
5.00	89	0.47	42.45
6.00	99	0.49	41.87

Note that the x column of each time step sample has a value close to 0.5, with some variability. If we then fit a time series with covariates to the condensed dataset, we obtain results with empirical means displayed in the table below. The “Significance Frequency” column displays the fraction of the hundred simulations in which the corresponding parameter estimate was statistically significant at the $\alpha = .05$ level.

Parameter	True Value	Estimate Mean	Sig. Freq.
phi	0.500	0.494	1.000
theta	0.500	0.001	0.040
Intercept	43.000	43.017	1.000
Slope	-0.001	-0.001	0.860
Covariate	0.250	0.216	0.110

Note that the effect of the dummy indicator variable is deemed significant only 11% of the time. This example highlights the shortcomings of fitting a time series

with covariates. Also of note is how skewed the sample mean of $\hat{\theta}$ is, in particular how close to zero it is. We will see later why this is actually expected. For now, we use these results to motivate the need for a different approach. With that need in mind, we next review some of the key developments in modeling repeated cross-sectional data.

3.2 Early Methods for Analyzing RCS Data

Among the earliest methods designed for RCS data was that proposed by Scott and Smith [16], in which dependent variables are aggregated at the time step level and modeled as a time series with added noise:

$$y_t = \theta_t + e_t, \quad \text{Var}(e_t) = s_t^2 \quad (3.2.1)$$

In this model, surveys are conducted at regular time intervals and θ_t represents a population mean changing over time. The analytical method proposed therein introduces different possibilities for the covariance structures of $\{\theta_t\}$ and $\{e_t\}$, but is ultimately concerned with the use of all previously observed aggregates $\{y_t\}_{t=1}^T$ to estimate θ_T . As our objective is to improve the estimation and of covariate effects and their standard errors, we must look to more highly parameterized models.

Such a model was developed by DiPrete and Grusky [4], in which added complexity and flexibility is added to a standard fixed effects model by having the effects of individual-level variables depend on macro-level variables:

$$y_{ti} = X_{ti}\beta_t + \varepsilon_{ti}, \quad \beta_t = \mathbf{Z}_t\gamma, \quad \varepsilon_{ti} \sim \text{WN}(0, \sigma^2) \quad (3.2.2)$$

Such a model requires the presence of macro-level variables that vary over time and explain changing effects of individual-level variables. While this formulation has a

broad range of applications, it is only appropriate in cases where temporal correlation can be assumed to be a function of known time-varying predictors, rather than an independent random process.

3.3 Autoregressive Random Effects Models

One model that has been developed to analyze RCS data with autocorrelated errors is the Autoregressive Random Effects (ARE) Model [10]. As it is the model that served as our starting point, we describe it in greater detail. Treating observations as the sum of a fixed effects model and an AR(1) time series, ARE models have two levels:

$$y_{ti} = \mathbf{x}_{ti}\beta + u_t + \epsilon_{ti}, \quad i = 1, \dots, n, \quad \epsilon_{ti} \sim \text{i.i.d. } N(0, \sigma_\epsilon^2) \quad (3.3.1)$$

$$u_t - \phi u_{t-1} = \eta_t + \theta \eta_{t-1}, \quad t = 1, \dots, T, \quad \eta_t \sim N(0, \sigma_\eta^2) \quad (3.3.2)$$

A realization of an ARE model is displayed in Figure 3.1 with the following parameters:

Parameter	β_0	β_t	β_1	σ_ϵ^2	ϕ	σ_η^2
Value	40	-0.1	2	1	0.9	1

Table 3.1: Parameter values for simulated ARE model

Setting $N := nT$, the total number of observations in the dataset, we define the fixed effect and temporal fluctuation design matrices X_f and X_t (resp.) to be:

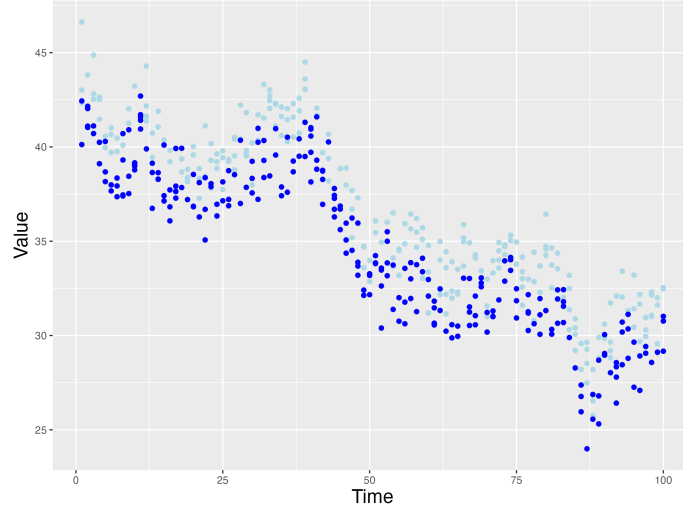


Figure 3.1: Realization of ARE model with parameters defined in Table 3.1

$$X_f = \begin{bmatrix} \mathbf{1}_n & \mathbf{1}_n & \mathbf{x}_1 \\ \mathbf{1}_n & \mathbf{2}_n & \mathbf{x}_2 \\ \vdots & \vdots & \vdots \\ \mathbf{1}_n & \mathbf{T}_n & \mathbf{x}_T \end{bmatrix}, \quad X_t = \begin{bmatrix} \ddots & & \mathbf{0} \\ & \mathbf{1}_n & \\ \mathbf{0} & & \ddots \end{bmatrix}$$

With these matrices we can concisely represent the model governing the vector of observations $\mathbf{y}$:

$$\mathbf{y} = X_f \beta + X_t \mathbf{u} + \boldsymbol{\varepsilon}$$

The method employed to fit such a model is a recursive process that splits the log likelihood function of the model into two terms and then utilizes an EM algorithm for parameter estimates. We introduce some necessary notation for expressing the likelihood function of the full set of parameters:

$$\mathbb{R}^{T \times T} \ni \Gamma := \text{Var}(\mathbf{u}), \quad \Gamma_{i,j} = \gamma_{\mathbf{u}}(i-j) = \frac{\phi^{|i-j|} \sigma_{\eta}^2}{1 - \phi^2} \quad (3.3.3)$$

Note that $\gamma_{\mathbf{u}}(h)$ always has a factor of σ_{η}^2 , meaning the Toeplitz matrix $\mathbf{V}$ can

be defined such that $\mathbf{\Gamma} = \sigma_\eta^2 \mathbf{V}$:

$$\mathbf{V} = \frac{1}{1 - \phi^2} \begin{bmatrix} 1 & \phi & \dots & \phi^{T-1} \\ \phi & 1 & \ddots & \phi^{T-2} \\ \vdots & \ddots & \ddots & \vdots \\ \phi^{T-1} & \phi^{T-2} & \dots & 1 \end{bmatrix} \quad (3.3.4)$$

We can also use X_t to concisely represent the covariance matrix of $\mathbf{y}$:

$$\mathbb{R}^{N \times N} \ni \Omega := \text{Var}(\mathbf{y}) = X_t \mathbf{\Gamma} X_t' + \sigma_\varepsilon^2 I_N \quad (3.3.5)$$

We define ξ to be the vector of all parameters $(\beta, \sigma_\varepsilon^2, \phi, \sigma_\eta^2)$. The vector $\mathbf{y}$ is a multivariate Gaussian random variable with mean $X_f \beta$, variance Ω , and the following log likelihood function:

$$l(\xi|\mathbf{y}) = \frac{-N}{2} \ln(2\pi) - \frac{1}{2} \ln|\Omega| - \frac{1}{2} (\mathbf{y} - X_f \beta)' \Omega^{-1} (\mathbf{y} - X_f \beta) \quad (3.3.6)$$

By conditioning on $\mathbf{u}$, we can split the log likelihood function into two levels:

$$l(\xi|\mathbf{y}) = l_1(\xi_1|\mathbf{y}, \mathbf{u}) + l_2(\xi_2|\mathbf{u}), \quad \xi_1 = (\beta, \sigma_\varepsilon^2), \quad \xi_2 = (\phi, \sigma_\eta^2)$$

$$l_1(\xi_1|\mathbf{y}, \mathbf{u}) = \ln(f_{Y|U}(\mathbf{y}|\mathbf{u}, \xi_1)) = -\frac{N}{2} \ln(2\pi\sigma_\varepsilon^2) - \frac{1}{2\sigma_\varepsilon^2} \|\mathbf{y} - X_t \mathbf{u} - X_f \beta\|^2 \quad (3.3.7)$$

$$l_2(\xi_2|\mathbf{u}) = \ln(f_U(\mathbf{u})) = -\frac{T}{2} \ln(2\pi\sigma_\eta^2) + \frac{1}{2} \ln(1 - \phi^2) - \frac{1}{2\sigma_\eta^2} \mathbf{u}' V^{-1} \mathbf{u} \quad (3.3.8)$$

This conditional splitting is convenient, as $\mathbf{u} \sim N(\mathbf{0}, \mathbf{\Gamma})$ and $\mathbf{y}|\mathbf{u} \sim N(X_f\beta + \mathbf{u}, \sigma_\varepsilon^2 I_N)$. Furthermore, the choice of imposing an AR(1) structure on $\mathbf{u}$ lends itself to a concise representation of the determinant and inverse of $\mathbf{\Gamma}$:

$$\mathbf{V}^{-1} = \begin{bmatrix} 1 & -\phi & 0 & \cdots & 0 & 0 \\ -\phi & 1 + \phi^2 & -\phi & \cdots & 0 & 0 \\ 0 & -\phi & 1 + \phi^2 & \ddots & 0 & 0 \\ \vdots & & \ddots & \ddots & -\phi & 0 \\ 0 & \cdots & 0 & -\phi & 1 + \phi^2 & -\phi \\ 0 & 0 & \cdots & & -\phi & 1 \end{bmatrix} \quad (3.3.9)$$

$$\mathbf{\Gamma}^{-1} = \sigma_\eta^{-2} \mathbf{V}^{-1}, \quad |\mathbf{\Gamma}| = \sigma_\eta^{2T} (1 - \phi^2) \quad (3.3.10)$$

In the absence of an explicit observation of $\mathbf{u}$, its estimate is repeatedly calculated and used to update parameter estimates. Its estimation is therefore a key part of this algorithm, and is given by its expectation, conditioned on $\mathbf{y}$:

$$\mathbb{E}(\mathbf{u}|\mathbf{y}) = \mathbf{\Gamma} X_t' \Omega^{-1} (\mathbf{y} - X_f \beta) \quad (3.3.11)$$

This estimate $\hat{\mathbf{u}}$ will have the following covariance structure:

$$\text{Var}(\hat{\mathbf{u}}) = \Sigma_{\hat{\mathbf{u}}, \hat{\mathbf{u}}} = \sigma_\eta^2 \mathbf{\Gamma} X_t' \Omega^{-1} X_t \mathbf{\Gamma}' \quad (3.3.12)$$

With this estimate now defined, we summarize the fitting of an ARE model:

- 1) Calculate initial parameter estimates;
- 2) Estimate $\mathbf{u}$ with $\mathbb{E}(\mathbf{u}|\mathbf{y})$;
- 3) Apply $\hat{\mathbf{u}}$ with an AR(1) likelihood to update $\hat{\xi}_2$ and fit $\mathbf{y} - X_t \hat{\mathbf{u}}$ with a fixed effects model to update $\hat{\xi}_1$;

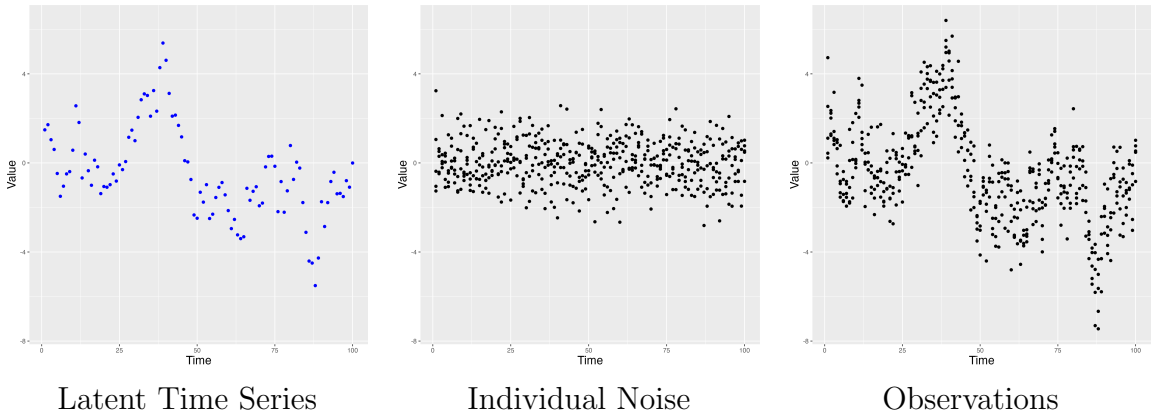
- 4) Calculate and store $l_1(\hat{\xi}_1)$ and $l_2(\hat{\xi}_2)$;
- 5) Repeat steps 2-4 until convergence criteria are met.

All estimate updates are based on explicitly calculable maximum likelihood estimators, with the exception of the AR parameter ϕ , which is updated via one iteration of the Newton-Raphson scheme since l_2 is not linear with respect to ϕ . In the next section, we examine some potential issues with fitting models with the ARE approach.

3.4 How Additional White Noise Alters ARMA Processes

Let's assume the best case scenario for level one of the model, namely the perfect estimation of fixed effect parameters ($\hat{\beta} = \beta$). In this unlikely event, we are able to take data such as that displayed in Figure 3.1 and extract $\mathbf{w} := X_t\mathbf{u} + \boldsymbol{\varepsilon}$, which is a realization of an *ARMA-Noise Model* [7]. This decomposition is displayed in Figure 3.2:

Figure 3.2: A realization of a repeatedly observed AR(1) time series with added noise



With no prior information about the latent time series, the best estimates for

the elements of $\mathbf{u}$ are the averages at each time step:

$$\hat{\mathbf{u}} = n^{-1} X_t'(X_t \mathbf{u} + \varepsilon) = \mathbf{u} + \mathbf{a} =: \bar{\mathbf{w}}, \quad a_t = n^{-1} \sum_{i=1}^n \varepsilon_{ti}$$

As is shown in Figure 3.3, the daily averages (in red) seem to follow the true latent time series (in blue) fairly closely. We will soon see, however, that the added noise vector $\mathbf{a}$ introduces bias in the estimation of the time series parameters. To describe this bias, let's compare the respective ACVF's of $\mathbf{u}$ and $\hat{\mathbf{u}}$.

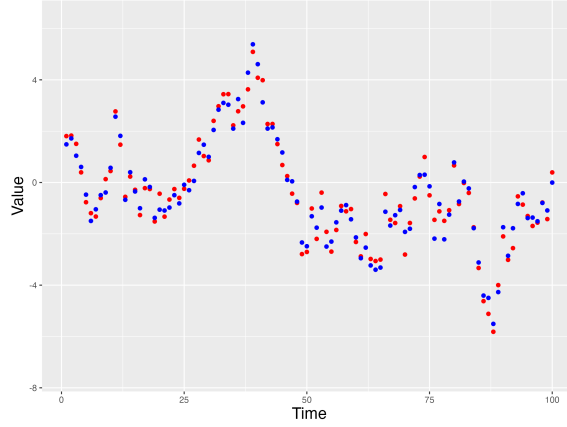


Figure 3.3: A Sample Latent Time Series (blue) and Daily Averages of Observed Data (red)

Being an AR(1) process, the covariance of u_t and $u_{t \pm h}$ is given by Equation 2.2.2. By virtue of the independence of $\{\eta_t\}$ and $\{a_t\}$, the ACVF of $\hat{\mathbf{u}}$ is affected as follows:

$$\gamma_{\hat{\mathbf{u}}}(h) = \begin{cases} \gamma_{\mathbf{u}}(0) + \frac{\sigma_{\varepsilon}^2}{n}, & h = 0 \\ \gamma_{\mathbf{u}}(h), & |h| > 0 \end{cases} \quad (3.4.1)$$

The fact that $\frac{\gamma_{\mathbf{u}+\mathbf{a}}(1)}{\gamma_{\mathbf{u}+\mathbf{a}}(0)} \neq \frac{\gamma_{\mathbf{u}+\mathbf{a}}(2)}{\gamma_{\mathbf{u}+\mathbf{a}}(1)}$ precludes $\hat{\mathbf{u}} = \mathbf{u} + \mathbf{a}$ from being an AR(1) process, but as we will now show, its ACVF is compatible with that of an ARMA(1,1). From Box (1976) [2], we have the following theorem:

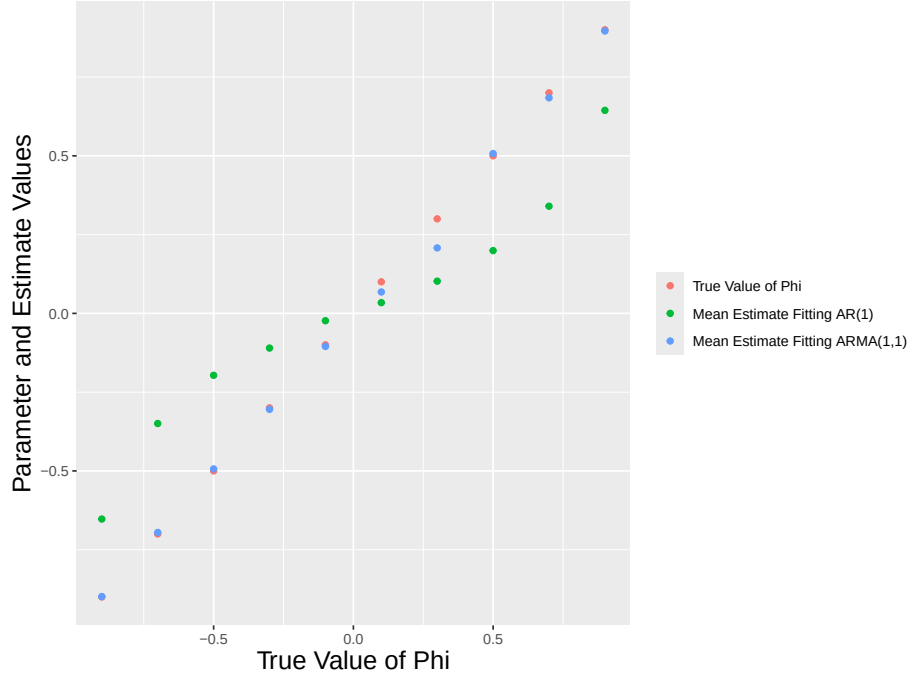


Figure 3.4: Average estimates for ϕ after fitting $\hat{\mathbf{u}} = \mathbf{u} + \mathbf{a}$ with an AR(1) model and an ARMA(1,1) model. The results were produced by 100 realizations with $T = 1000$, $\sigma_\eta^2 = \sigma_a^2$. The disparity between the accuracy of the two means grows with the relative magnitude of σ_a^2 with respect to σ_η^2 .

If $\{y_t\}$ follows an ARMA(p_1, q_1) model, $\{a_t\}$ follows an ARMA(p_2, q_2) and the two series are independent, then their sum $w_t = y_t + a_t$ follows an ARMA(p, q) model in which $p \leq p_1 + p_2$ and $q \leq \max(p_1 + q_2, p_2 + q_1)$.

Because $\{a_t\}$ is white noise, $p_2 = q_2 = 0$. As $p_1 = 1$ and $q_1 = 0$, we can deduce that $\{w_t\}$ is an ARMA(p, q) process where $p \leq 1$ and $q \leq 1$. Because $\gamma_{\hat{\mathbf{u}}}(h + 1)/\gamma_{\hat{\mathbf{u}}}(h) = \phi$ for $h > 0$, we know that $p = 1$. The ACVF of $\hat{\mathbf{u}}$ is therefore incompatible with that of an AR(1) process, so it must be ARMA(1,1).

The fact that both ACVF's are altered by a factor of ϕ after lag 1 means that the AR component is unaffected by the addition of white noise. While noting the AR term ϕ goes unchanged by the added noise, the estimation thereof will be irredeemably biased if fit with an AR(1) model (See Figure 3.4). $\xi_2 = (\phi, \sigma_\eta^2)$ must instead be estimated with a method that accounts for this additional noise

3.5 Analyzing ARMA-Noise Processes

Remaining in our best-case scenario of perfectly estimating β , our objective is to fit estimates for $\phi, \theta, \sigma_\eta^2, \sigma_\varepsilon^2$ with observations $\{w_{ti}\}$, the time-step averages of which form a time series $\bar{\mathbf{w}}$ that is parameterized by $\phi, \theta_b, \sigma_b^2$ (with innovations $\{b_t\}$). ARMA-Noise models are fit with a method detailed by Wong and Miller (1990) [7], summarized as follows. We note that true parameters of the model can only be effectively modeled if $n > 1$, as it enables the variance σ_ε^2 to be teased out and estimated as follows:

$$e \tag{3.5.1}$$

The fact that $\mathbf{u}$ and $\mathbf{w}$ share a common AR parameter means that we also can obtain a reliable estimator for ϕ by fitting an ARMA(1,1) model to $\mathbf{w}$. In doing so, we will also obtain estimates for θ_b and σ_b^2 , which we can use to obtain an estimate for our missing parameter of interest, σ_η^2 . Let us start by using the backwards operator B to rewrite the equation $\bar{w}_t = u_t + a_t$:

$$(1 - \phi B)^{-1}(1 + \theta_b B)b_t = (1 - \phi B)^{-1}\eta_t + a_t \tag{3.5.2}$$

Multiplying the above equation by $(1 - \phi B)$ yields:

$$(1 + \theta_b B)b_t = \eta_t + (1 - \phi B)a_t \tag{3.5.3}$$

By virtue of the independence of η_r and a_s for any $r, s \in \mathbb{Z}$, we now have an equation that resembles white noise and an MA(1) process on the right and a new MA(1) process on the left. We can then relate the parameters involved on the left and right sides using their respective ACVF's at lag 0,

$$(1 + \theta_b^2)\sigma_b^2 = \sigma_\eta^2 + (1 + \phi^2)\sigma_a^2 \quad (3.5.4)$$

We will set this system of equations equal to 0 and call it H_θ :

$$H_\theta = (1 + \theta_b^2)\sigma_b^2 - \sigma_\eta^2 - (1 + \phi^2)\sigma_a^2 \quad (3.5.5)$$

$H_\theta : \mathbb{R}^4 \rightarrow \mathbb{R}$ is now a function that implicitly defines $\hat{\sigma}_\eta^2$ as the root of $H_\theta(\hat{\phi}, \hat{\theta}_b, \hat{\sigma}_b^2, \hat{\sigma}_a^2)$. We then define $\hat{\sigma}_a^2 := n^{-1}\hat{\sigma}_\varepsilon^2$. While it may seem a trivial matter to define an explicit function for σ_η^2 , we will maintain the implicit nature of H for the purposes of generalizing later. Our estimates for ϕ and σ_η^2 will now be consistent, with asymptotic distributions to be described in the Methods chapter.

3.5.1 Generalizing ARMA Order of u

With reasoning similar to that used in the previous section, we can establish that the sum of an ARMA(1,1) process and white noise is an ARMA process with $p = 1$ and $q \leq 1$. Let's see how Equation 3.5.4 is affected:

$$(1 - \phi B)^{-1}(1 + \theta_b B)b_t = (1 - \phi B)^{-1}(1 + \theta B)\eta_t + a_t \quad (3.5.6)$$

We multiply both sides by $(1 - \phi B)$ as before:

$$(1 + \theta_b B)b_t = (1 + \theta B)\eta_t + (1 - \phi B)a_t \quad (3.5.7)$$

Now a relation between three MA(1) process, the respective ACVF's at lags 0 and 1 produce the following system of equations:

$$\begin{cases} (1 + \theta_b^2)\sigma_b^2 = (1 + \theta^2)\sigma_\eta^2 + (1 + \phi^2)\sigma_a^2 \\ \theta_b\sigma_b^2 = \theta\sigma_\eta^2 - \phi\sigma_a^2 \end{cases} \quad (3.5.8)$$

Setting both equations equal to 0 thus defines our more general form for H . While the system of equations has a quadratic equation for θ , empirical results indicate that one solution is always of absolute value greater than 1, meaning only one of the two solutions is viable.

$$H = \begin{cases} (1 + \theta_b^2)\sigma_b^2 - (1 + \theta^2)\sigma_\eta^2 - (1 + \phi^2)\sigma_a^2 \\ \theta_b\sigma_b^2 - \theta\sigma_\eta^2 + \phi\sigma_a^2 \end{cases} \quad (3.5.9)$$

Since this change in the order of $\mathbf{u}$ doesn't change the maximum order of $\bar{\mathbf{w}}_t$, we generalize $\mathbf{u}$ in our model to be an ARMA(1,1) process. Figures 3.5-3.7 shed light on how added noise alters the MA component of ARMA(1,1) processes. Note that in both Figures 3.5 and ?? all curves corresponding to a value of θ of the same sign as ϕ eventually cross the x-axis, suggesting that specific parameter combinations of $\phi, \theta, \sigma_a^2, \sigma_\eta^2$ result in $\bar{\mathbf{w}}_t$ an AR(1) process as opposed to one that is ARMA(1,1), but we will go into further detail in later sections.

In Chapter 4 we incorporate this method for fitting a model with fixed effects. We also detail how inference is performed on resulting estimates. Because of its increased capacity, we generalize the class of applicable models to those that are the sum of a fixed effects model and ARMA(1,1) process. However, even if the previously proposed EM algorithm included an MA parameter, it would still be estimating $\hat{\beta}$ and σ_ε^2 by treating $\mathbf{y} - X_t\hat{\mathbf{u}}$ as a realization of a fixed effects model with white noise in the residuals. We now examine the implications of such a treatment.

3.6 Impacts of time series estimation error on confidence regions for regression parameters

We've already looked at the implications of perfectly estimating β to improve how Level 2 parameters are estimated. Let's now go in the other direction, assuming

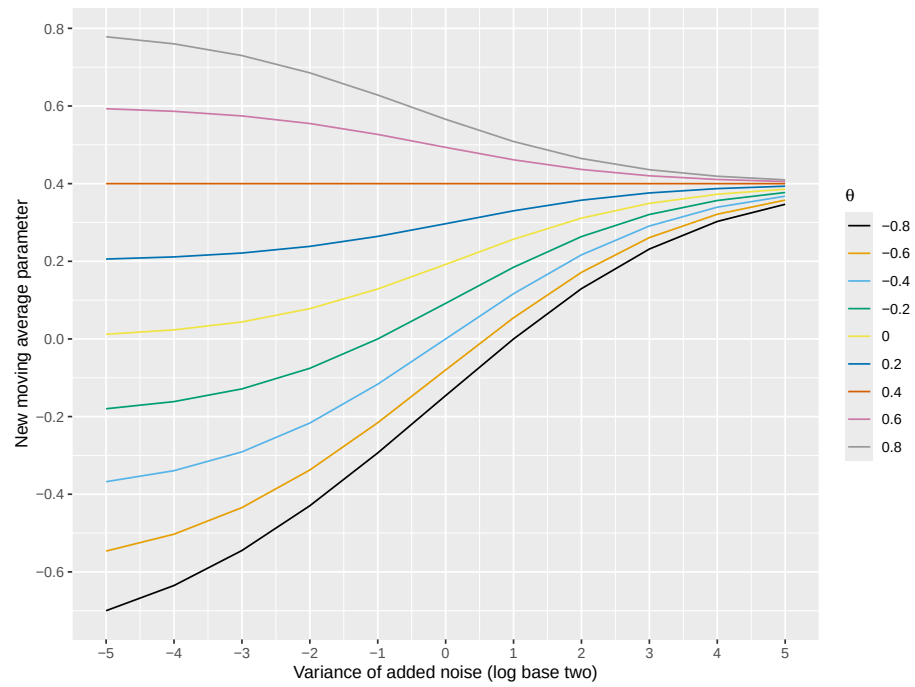


Figure 3.5: The effect of noise on the MA parameter of ARMA(1,1) processes with $\phi = -0.4$

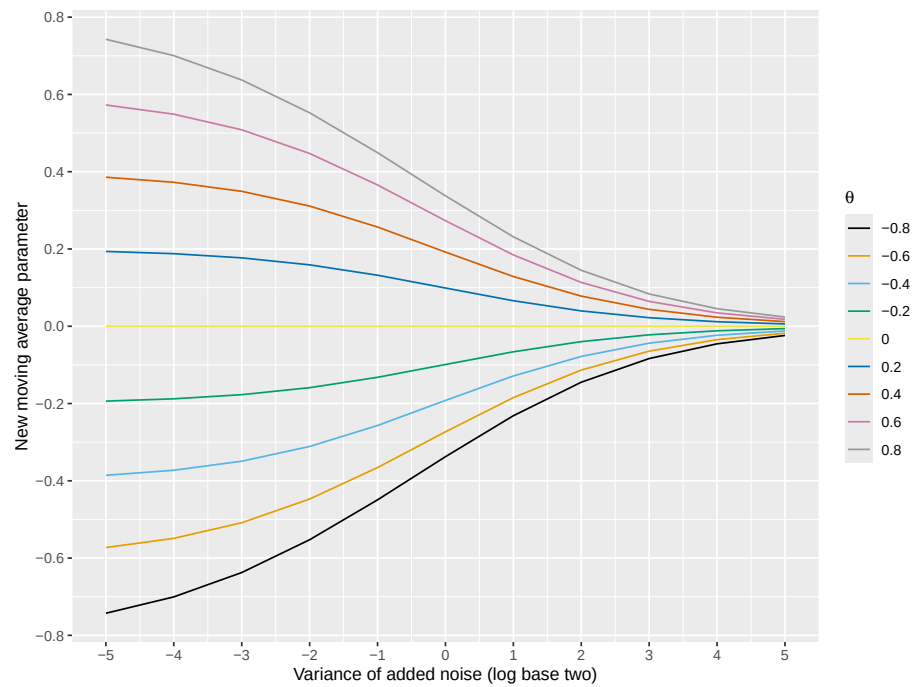


Figure 3.6: The effect of noise on the MA parameter of MA(1) processes

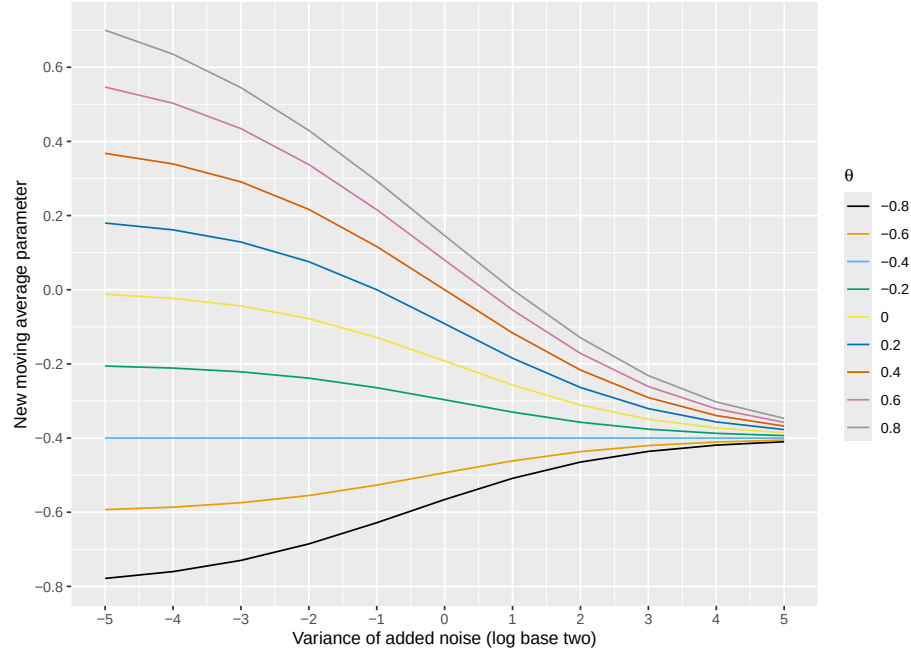


Figure 3.7: The effect of noise on the MA parameter of ARMA(1,1) processes with $\phi = 0.4$

perfect estimation of ξ_2 to investigate the implications on fitting $\hat{\beta}$. This hypothetical is somewhat contrived, as we could only isolate $X_t \mathbf{u} + \varepsilon$ in the event that β was estimated perfectly, in which case we would have no need to estimate it again. To that end, let's assume that our algorithm of choice perfectly estimates β on a particular update, but doesn't "know" it. It then perfectly fits $\hat{\xi}_2$ using the method outlined by Wong, and the question becomes whether β is a "stable" estimate.

In the event that ξ_2 is perfectly estimated, all parameter estimates $\hat{\xi}$ are plugged into the conditional expectation formula given in Equation 3.3.11. Even with perfect parameter estimates, $\hat{\mathbf{u}}$ will still deviate from $\mathbf{u}$, as $\text{Var}(\hat{\mathbf{u}} - \mathbf{u}) = \mathbf{\Gamma} - \mathbf{\Gamma} X_t' \Omega^{-1} X_t \mathbf{\Gamma}$. As a result, $\widehat{X_f \beta} + \varepsilon = \mathbf{y} - X_t \hat{\mathbf{u}}$ will not be the sum of fixed effects and white noise as the current algorithm supposes. Were we to treat $\mathbf{y} - X_t \hat{\mathbf{u}}$ as such, our resulting estimate for β would be the standard OLS estimate $\hat{\beta} = (X_f' X_f)^{-1} X_f' (\mathbf{y} - X_t \hat{\mathbf{u}})$

However, the inevitable error in $\hat{\mathbf{u}}$ would endow $\hat{\beta}$ not with the implied $\text{Var}(\hat{\beta}) = (X_f' X_f)^{-1} \sigma_\varepsilon^2$, but rather the following covariance matrix:

$$\text{Var}(\hat{\beta}) = 2\sigma_\varepsilon^2 (X_f' X_f)^{-1} - \sigma_\varepsilon^2 (X_f' X_f)^{-1} X_f' (\Omega^{-1}) X_f (X_f' X_f)^{-1} =: \Sigma_\beta \quad (3.6.1)$$

As convenient as it might be to subtract corresponding entries of $\hat{\mathbf{u}}$ from $\mathbf{y}$ and treat the resulting difference as a fixed effects model with independent noise, the resulting confidence regions would be incorrect.

As the best-case scenario of estimating and extracting the latent time series will still yield temporal correlation in the data, albeit reduced, we must turn to a more nuanced treatment of residual covariance. A natural choice is the GLS estimator of β , which we will see challenges the necessity of using $\hat{\mathbf{u}}$ in the estimation of β at all.

As we now have a viable method (courtesy of Wong and Miller) for estimating all the parameters necessary to estimate Ω , an analog of the above equation with sample estimates will be our estimator for β , and confidence intervals will be based on a sample estimate of its covariance matrix $\text{Var}(\hat{\beta}_{\text{GLS}}) = (X_f' \Omega^{-1} X_f)^{-1}$. With all these considerations in mind, we are now ready to propose our method for fitting these types of models.

Chapter 4

Methods

4.1 Incorporating ARMA-Noise into EM Algorithm

Now that the analysis of ARMA-Noise processes have been discussed, we examine how such an analysis may be applied, not to explicit realizations of ARMA-Noise processes, but rather estimates of one in the form of residuals from a fixed effects model. We define our model similarly to the ARE model, but with the latent time series $\mathbf{u}$ now specified as an ARMA(1,1) model as opposed to strictly AR(1):

$$y_{ti} = \mathbf{x}_{ti}\beta + u_t + \epsilon_{ti}, \quad i = 1, \dots, n \quad \epsilon_{ti} \sim \text{i.i.d. } N(0, \sigma_\epsilon^2) \quad (4.1.1)$$

$$u_t - \phi u_{t-1} = \eta_t + \theta \eta_{t-1}, \quad t = 1, \dots, T \quad \eta_t \sim N(0, \sigma_\eta^2) \quad (4.1.2)$$

The algorithm we propose is an iterative process applying the estimates of parameters from one level to update those of the other. The process begins similarly

to that outlined by Modugno *et al*, with a few key differences:

- 1) Estimate β with $\hat{\beta} = (X_f'X_f)^{-1}X_f'\mathbf{y}$;
- 2) Estimate $X_t\mathbf{u} + \varepsilon$ with $\mathbf{y} - X_f\hat{\beta}$ and fit ARMA-Noise model to obtain estimates $(\hat{\phi}, \hat{\theta}, \hat{\sigma}_\eta^2, \hat{\sigma}_\varepsilon^2)$;
- 3) Calculate $\hat{\Omega}$ and $l(\hat{\xi})$;
- 4) Refit fixed effect parameter estimates by calculating GLS estimate $\hat{\beta}_{GLS} = (X_f'\hat{\Omega}^{-1}X_f)X_f'\hat{\Omega}^{-1}\mathbf{y}$;
- 5) Repeat Steps 2-4 until convergence criteria are met.;
- 6) Generate confidence regions and eliminate insignificant parameters if applicable.

4.1.1 Evaluating Convergence

Instead of splitting the likelihood by conditioning on $\hat{\mathbf{u}}$, doing so with $\bar{\mathbf{y}}$ instead of has the advantage that likelihoods are based on a constant statistic, as opposed to the ever-changing estimate of $\mathbf{u}$.

Let's recall the log likelihood $l(\xi)$:

$$l(\xi) = -\frac{N}{2}\ln(2\pi) - \frac{1}{2}\ln|\Omega| - \frac{1}{2}(\mathbf{y} - X_f\beta)'\Omega^{-1}(\mathbf{y} - X_f\beta)$$

We now condition on $\bar{\mathbf{y}}_t = n^{-1}X_t'\mathbf{y}$, which has the following distribution

$$\bar{\mathbf{y}} \sim N(n^{-1}X_t'X_f\beta, n^{-2}X_t'\Omega X_t)$$

After simplifying, we get the following expression for $\Sigma_{\bar{\mathbf{y}}}$:

$$\Sigma_{\bar{\mathbf{y}}} = \Gamma + n^{-1}\sigma_\varepsilon^2 I_T,$$

which has the following log likelihood:

$$l_t(\xi) = \ln[f(\bar{\mathbf{y}}|\xi)] = -\frac{T}{2}\ln(2\pi) - \frac{1}{2}\ln|\Sigma_{\bar{\mathbf{y}}}| - \frac{1}{2}(\bar{\mathbf{y}} - n^{-1}X_t'X_f\beta)' \Sigma_{\bar{\mathbf{y}}}^{-1}(\bar{\mathbf{y}} - n^{-1}X_t'X_f\beta)$$

If we then condition $\mathbf{y}$ on $\bar{\mathbf{y}}$, the only randomness left is in the vector of deviations, which are independent of $\bar{\mathbf{y}}$:

$$\mathbf{y} - X_t\bar{\mathbf{y}} = \boldsymbol{\varepsilon} - X_t\mathbf{a}$$

Let's establish the distribution of this vector of deviations. The mean is immediately zero, and the covariance is given by:

$$\text{Var}(\boldsymbol{\varepsilon} - X_t\mathbf{a}) = \text{Var}(\boldsymbol{\varepsilon}) - 2\text{Cov}(\boldsymbol{\varepsilon}, X_t\mathbf{a}) + \text{Var}(X_t\mathbf{a})$$

which reduces to :

$$\text{Var}(\boldsymbol{\varepsilon} - X_t\mathbf{a}) = \sigma_\varepsilon^2 I_N - 2\sigma_\varepsilon^2 n^{-1} X_t X_t' + n^{-2} \sigma_\varepsilon^2 (X_t X_t')^2$$

Because $X_t'X_t = nI_N$, we get:

$$\text{Var}(\boldsymbol{\varepsilon} - X_t\mathbf{a}) = \sigma_\varepsilon^2 (I_N - n^{-1} X_t X_t') \quad (4.1.3)$$

We now derive the second part of our split likelihood. Because of the rank-deficient nature of the matrix Equation 4.1.3, we can simplify the log likelihood

$$l_i(\xi) = \ln f(\mathbf{y} \mid \bar{\mathbf{y}}, \sigma_\varepsilon^2) = -\frac{N-T}{2} \ln(2\pi\sigma_\varepsilon^2) - \frac{1}{2\sigma_\varepsilon^2} \sum_{t=1}^T \sum_{i=1}^n (y_{it} - \bar{y}_t)^2, \quad (4.1.4)$$

With this we can express the full likelihood as $l(\xi) = l_t(\xi) + l_i(\sigma_\varepsilon^2)$. However, since $\bar{\mathbf{y}}_t$ is independent of our model selection and parameter estimates, l_i is going to be relatively stable. We therefore pair $\xi \in \mathbb{R}^k$ and $l_t(\xi)$ to calculate the Akaike Information Criterion (AIC):

$$AIC = -2k - 2l_t(\xi)$$

Calculating this weighted likelihood is how we will compare the performance of models that include different subsets of parameters.

4.2 Inference

The equation H provides not only the information necessary for the estimation of $\tau_l = (\phi, \theta, \sigma_\eta^2, \sigma_\varepsilon^2)$, but also the asymptotic distribution of $\hat{\tau}_l$, via a combination of the Delta Method and the Implicit Function Theorem [7]. Recall from Section 2 the asymptotic distribution of the normalized estimate of $\tau_o = (\phi, \theta_b, \sigma_b^2, \sigma_\varepsilon^2)$:

$$\sqrt{T} \left[\begin{pmatrix} \hat{\phi} \\ \hat{\theta}_b \\ \hat{\sigma}_b^2 \end{pmatrix} - \begin{pmatrix} \phi \\ \theta_b \\ \sigma_b^2 \end{pmatrix} \right] \xrightarrow{d} N(\mathbf{0}, \Sigma_b)$$

$$\Sigma_b = \begin{bmatrix} \frac{(1-\phi^2)(1+\phi\theta_b)^2}{(\phi+\theta_b)^2} & \frac{-(1-\theta_b^2)(1-\phi^2)(1+\phi\theta_b)}{(\phi+\theta_b)^2} & 0 \\ \frac{-(1-\theta_b^2)(1-\phi^2)(1+\phi\theta_b)}{(\phi+\theta_b)^2} & \frac{(1-\theta_b^2)(1+\phi\theta_b)^2}{(\phi+\theta_b)^2} & 0 \\ 0 & 0 & 2\sigma_b^4 \end{bmatrix}$$

We also have the asymptotic distribution of the normalized estimate for σ_ε^2 :

$$\sqrt{T}(\hat{\sigma}_\varepsilon^2 - \sigma_\varepsilon^2) \xrightarrow{d} N(0, \frac{2\sigma_\varepsilon^4}{n-1})$$

Because $y_t \perp S_t^2$ for all t , we know that $\hat{\sigma}_\varepsilon^2 \perp (\hat{\phi}, \hat{\theta}_b, \hat{\sigma}_\eta^2)$ and can give the

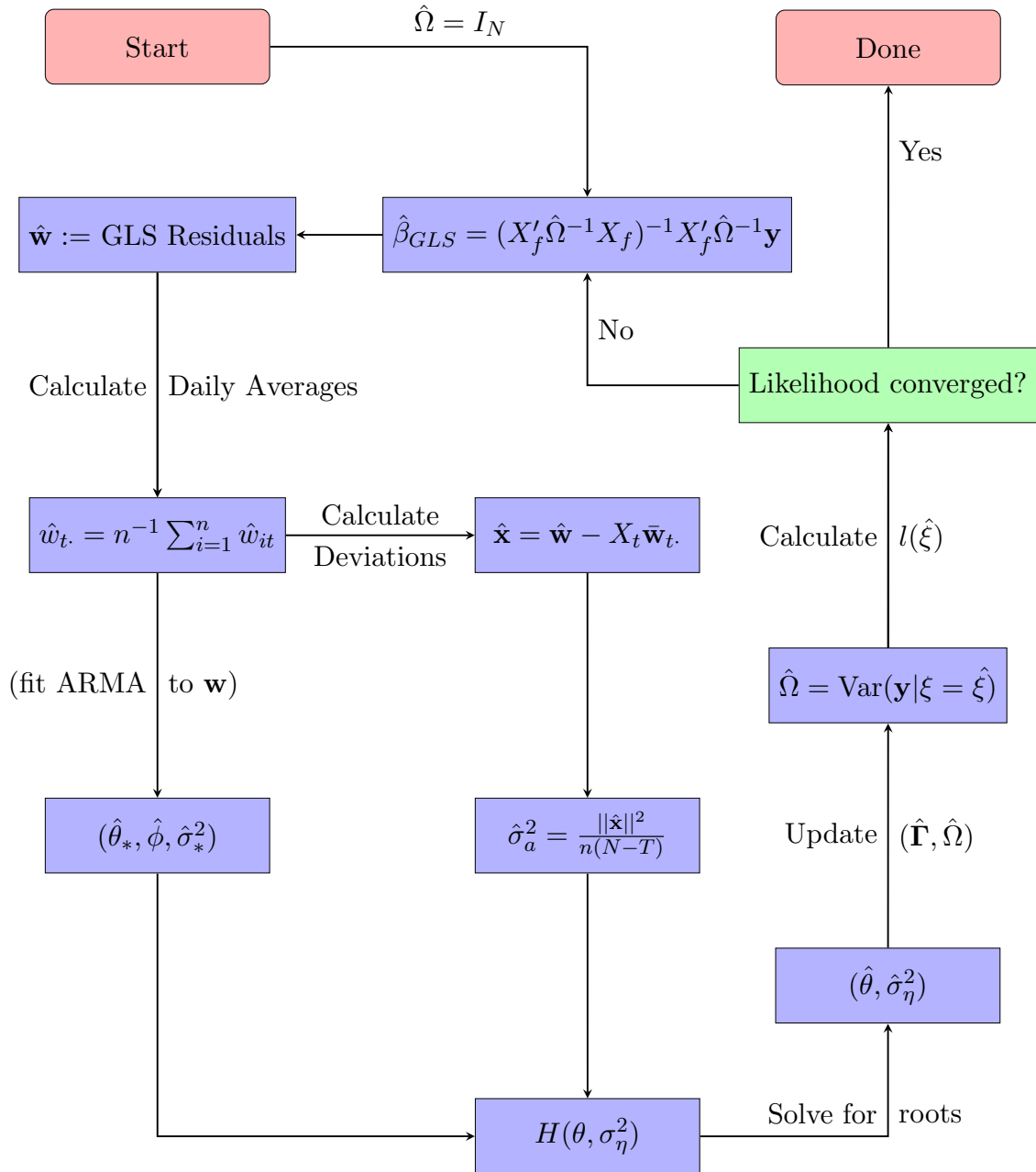


Figure 4.1: Flowchart of algorithm steps for estimating model parameters

following asymptotic distribution:

$$\sqrt{T}(\hat{\tau}_o - \tau_o) \xrightarrow{d} N(\mathbf{0}, \Sigma_o), \quad \Sigma_o = \begin{bmatrix} \Sigma_b & \mathbf{0} \\ \mathbf{0} & \frac{2\sigma_\varepsilon^2}{n-1} \end{bmatrix} \quad (4.2.1)$$

Equipped with the above estimates, we would then solve H for estimates of θ and σ_η^2 . It is important to note that estimates for τ_o can produce a system of equations that has no real roots, especially for relatively low values of T . In this event we use numerical methods to solve for parameter estimates $\hat{\theta}, \hat{\sigma}_\eta^2$ that minimize $\|H\|^2$.

Were we to have an explicit function $(\hat{\phi}, \hat{\theta}, \hat{\sigma}_\eta^2) = F(\hat{\tau}_o)$ providing estimates for τ_l , we would apply the Delta Method to obtain the asymptotic distribution of our vector of parameter estimates:

$$\sqrt{T}(F(\hat{\tau}_o) - \tau_l) \xrightarrow{d} N(\mathbf{0}, G\Sigma_o G'), \quad G = \frac{\partial F}{\partial(\boldsymbol{\tau}_o)} \quad (4.2.2)$$

Because we're using H , a function whose roots are our estimates, we will adapt 4.2.2 using the Implicit Function Theorem:

$$G = - \left(\frac{\partial H_f}{\partial \boldsymbol{\tau}_1} \right)^{-1} \left(\frac{\partial H}{\partial \boldsymbol{\tau}_0} \right) \quad (4.2.3)$$

Because we're ultimately interested in the normalized asymptotic distribution of parameter estimates $(\hat{\phi}, \hat{\theta}, \hat{\sigma}_\eta^2, \hat{\sigma}_\varepsilon^2)$, we expand H to include some identity functions.

$$H_f : \begin{cases} \phi_* - \phi \\ \sigma_{\varepsilon*}^2 - \sigma_\varepsilon^2 \\ (1 + \theta^2)\sigma_\eta^2 + (1 + \phi^2)\sigma_\varepsilon^2/n - (1 + \theta_b^2)\sigma_b^2 \\ \theta\sigma_\eta^2 - \phi\sigma_\varepsilon^2/n - \theta_b\sigma_b^2 \end{cases} \quad (4.2.4)$$

Our procedure is to fit $\hat{\tau}_o$, solve for $\hat{\tau}_l$, and then plug in all estimates into Equations 4.2.2 and 4.2.3

$$\tau_0 = \begin{pmatrix} \phi \\ \theta_b \\ \sigma_b^2 \\ \sigma_\varepsilon^2 \end{pmatrix} \quad \tau_1 = \begin{pmatrix} \phi_* \\ \sigma_{\varepsilon*}^2 \\ \theta \\ \sigma_\eta^2 \end{pmatrix} \quad (4.2.5)$$

In verifying that the estimation of these parameters proceeds as expected, we can utilize the distribution of the following statistic:

$$T(\hat{\tau}_o - \tau_o)' \Sigma_o^{-1} (\hat{\tau}_o - \tau_o) \stackrel{d}{=} \chi_3^2 \quad (4.2.6)$$

We can therefore test whether estimates of the parameters of an ARMA-Noise process fall within predicted confidence regions for known parameters.

4.3 Model Selection

Before outlining the methodology for model selection, we motivate the procedure by exploring the implications of setting certain parameters in an ARMA-Noise model $\bar{\mathbf{w}}_t = \mathbf{u} + \mathbf{a}$ equal to zero.

Case 1: $\mathbf{u}$ is AR(1)

We have already shown that this case results in $\mathbf{w}$ being an ARMA(1,1) process with innovations $\{b_t\}$ and parameterized by $(\phi, \theta_b, \sigma_b^2)$. Figures 3.5 and 3.7 remind us that θ_b will fall somewhere between 0 and $-\phi$, converging to the former as $\frac{\sigma_a^2}{\sigma_\eta^2}$ approaches 0 and the latter as it approaches infinity.

Assuming values of ϕ , σ_a^2 , and T of large enough magnitude to result in significant estimates for ϕ and θ_b , and not knowing $\mathbf{u}$ is AR(1), we use Equations 3.5.9, 4.2.2, and 4.2.3 to estimate and perform inference on $\hat{\tau}_l$. In the event $\hat{\theta}$ is insignificant

we impose $\theta = 0$ and re-estimate σ_η^2 using H_θ .

$$H_\theta = (1 + \theta_b^2)\sigma_b^2 - \sigma_\eta^2 - (1 + \phi^2)\sigma_a^2, \quad H_f = \begin{cases} \sigma_{\varepsilon*}^2 - \sigma_\varepsilon^2 \\ \phi_* - \phi \\ H_\theta \end{cases} \quad (4.3.1)$$

$$\Sigma_o = \begin{bmatrix} \Sigma_{\phi, \theta_b} & \mathbf{0} & \mathbf{0} \\ \mathbf{0}' & 2\sigma_b^4 & 0 \\ \mathbf{0}' & 0 & \frac{2\sigma_\varepsilon^4}{n-1} \end{bmatrix} \quad (4.3.2)$$

$$\tau_o = (\phi, \theta_b, \sigma_b^2, \sigma_\varepsilon^2)', \quad \tau_l = (\sigma_{\varepsilon*}^2, \phi_*, \sigma_\eta^2)'$$

$$\frac{\partial H_f}{\partial \tau_o} = \begin{bmatrix} 0 & 0 & 0 & -1 \\ -1 & 0 & 0 & 0 \\ -2\phi\sigma_a^2 & 2\theta_b\sigma_b^2 & (1 + \theta_b^2) & (1 + \phi^2)n^{-1} \end{bmatrix}, \quad \frac{\partial H_f}{\partial \tau_l} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

Case 2: $\mathbf{u}$ is MA(1)

We now consider the case where $\phi = 0$. If $\mathbf{u}$ is MA(1), so too is $\bar{\mathbf{w}}_t$, with parameters θ_b and σ_b^2 . The resulting system of related ACVF's is:

$$(1 + \theta_b B)b_t = (1 + \theta B)\eta_t + a_t$$

With this restricted relation we adapt our implicit function:

$$H_\phi = \begin{cases} (1 + \theta_b^2)\sigma_b^2 - (1 + \theta^2)\sigma_\eta^2 - \sigma_a^2 \\ \theta_b\sigma_b^2 - \theta\sigma_\eta^2 \end{cases}$$

Because the added noise in no way alters the lack of auto-regression, fitting an ARMA(1,1) process to $\mathbf{w}$ should yield an insignificant estimate for ϕ . In this event, $\mathbf{w}$ would be refit with an MA(1) model before estimating θ and σ_η^2 with an adjusted version of H , which we will denote H_ϕ :

$$H_\phi = \begin{cases} (1 + \theta_b^2)\sigma_b^2 - (1 + \theta^2)\sigma_\eta^2 - \sigma_a^2 \\ \theta_b\sigma_b^2 - \theta\sigma_\eta^2 \end{cases} \quad (4.3.3)$$

$$\Sigma_o = \begin{bmatrix} 1 - \theta^2 & 0 & 0 \\ 0 & 2\sigma_b^2 & 0 \\ 0 & 0 & \frac{2\sigma_\varepsilon^2}{n-1} \end{bmatrix} \quad (4.3.4)$$

$$\tau_o = (\theta_b, \sigma_b^2, \sigma_\varepsilon^2)' \quad \tau_l = (\sigma_\varepsilon^2, \theta, \sigma_\eta^2)'$$

$$H_f = \begin{bmatrix} \sigma_\varepsilon^2 - \sigma_{\varepsilon*}^2 \\ H_\phi \end{bmatrix}$$

$$\frac{\partial H_f}{\partial \tau_o} = \begin{bmatrix} 0 & 0 & -1 \\ 2\theta_b\sigma_b^2 & 1 + \theta_b^2 & -n^{-1} \\ \sigma_b^2 & \theta_b & 0 \end{bmatrix}, \quad \frac{\partial H_f}{\partial \tau_l} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -2\theta\sigma_\eta^2 & -(1 + \theta^2) \\ 0 & -\sigma_\eta^2 & -\theta \end{bmatrix}$$

Case 3: $\theta_b = 0$

We now turn to a case that was hinted at by Figures 3.5 and 3.7, namely that $\bar{\mathbf{w}}$ is AR(1). We've already established the sum of MA(1) and white noise is still MA(1). This means that $\mathbf{u}$ must be ARMA(1,1), setting up the following system of ACVF's:

$$(1 - \phi B)^{-1} b_t = (1 - \phi B)^{-1} (1 + \theta B) \eta_t + a_t$$

$$b_t = (1 + \theta B) \eta_t + (1 - \phi B) a_t$$

We calculate respective ACVF's at lags 0 and 1 to obtain:

$$H_{\theta_b} = \begin{cases} \sigma_b^2 - (1 + \theta^2) \sigma_\eta^2 - (1 + \phi^2) \sigma_a^2 \\ \theta \sigma_\eta^2 - \phi \sigma_a^2 \end{cases} \quad (4.3.5)$$

Note that the second equation provides a critical piece of information, namely that θ_b being zero necessitates that $\frac{\theta}{\phi} = \frac{\sigma_a^2}{\sigma_\eta^2}$. This gives rise to the following:

$$\Sigma_o = \begin{bmatrix} 1 - \phi^2 & 0 & 0 \\ 0 & 2\sigma_b^4 & 0 \\ 0 & 0 & \frac{2\sigma_\varepsilon^4}{n-1} \end{bmatrix} \quad (4.3.6)$$

$$\tau_o = (\phi, \sigma_b^2, \sigma_\varepsilon^2)', \quad \tau_l = (\sigma_{\varepsilon*}^2, \phi_*, \theta, \sigma_\eta^2), \quad H_f = \begin{cases} \sigma_{\varepsilon*}^2 - \sigma_\varepsilon^2 \\ \phi_* - \phi \\ H_{\theta_b} \end{cases}$$

$$\frac{\partial H_f}{\partial \tau_o} = \begin{bmatrix} 0 & 0 & -1 \\ -1 & 0 & 0 \\ -2\phi\sigma_a^2 & 1 & -(1 + \phi^2)\hat{m} \\ -\sigma_a^2 & 0 & -\phi\hat{m} \end{bmatrix}, \quad \frac{\partial H_f}{\partial \tau_l} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -2\theta\sigma_\eta^2 & -(1 + \theta^2) \\ 0 & 0 & \sigma_\eta^2 & \theta \end{bmatrix}$$

In the event of an insignificant estimate for θ_b , we would refit $\mathbf{w}$ with an AR(1) process and then use H_{θ_b} to fit the remaining parameters. We synthesize the implications of these cases with the following procedure for model selection, which is performed using the parameter estimates of the converged REMARCS model.

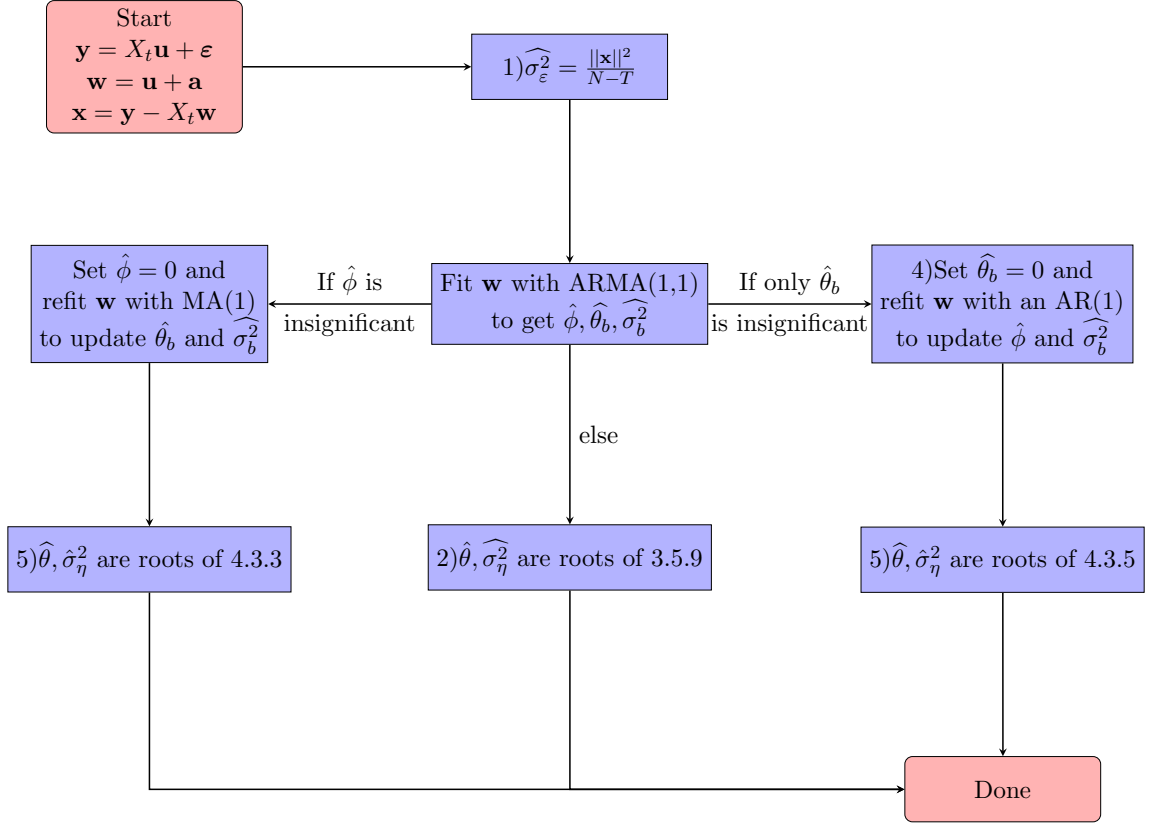


Figure 4.2: Decision tree for selecting parameters to include in ARMA-Noise model

Chapter 5

Varying Sample Sizes

When framed in the context of repeated surveys, the requirement that the sample size at each time step is constant is unrealistic. Given that real life survey data is rarely so uniform, generalizing the method to allow for varying daily sample size is a natural goal. In examining the method as it stands, the condition of fixed sample size is only necessary in the fitting of ARMA-Noise models, as $\mathbf{w} = \mathbf{u} + \mathbf{a}$ is only an ARMA process if $\{a_t = n_t^{-1} \sum_{i=1}^{n_t}\}$ is white noise, something that at first glance is violated by varying n_t . However, we will now show how this method can be adapted to allow for varying sample sizes, so long as certain conditions are met.

5.1 Time-Step Averages of White Noise with Varying Sample Size

We seek to take the ARMA-Noise model and generalize it to allow for $\{n_t\}$ to vary, as is so often the case in real RCS survey data. We will now treat $\{n_t\}$ as the realizations of random variables $\{N_t\}_{t=1}^T$. What we aim to establish are the conditions under which the random variable $A_t = N_t^{-1} \sum_{i=1}^{N_t} \varepsilon_{ti}$ is still white noise.

First, let $\{N_t\}$ be i.i.d. random variables with a sample space that is a subset

of the integers strictly greater than 0. The conditional random variable $A_t|N_t$ then has variance σ_ε^2/N_t . However, note that this is the variance of A_t after being conditioned on N_t , not its marginal variance.

To find $\text{Var}(A)$ we use the Law of Total Variation:

$$\text{Var}(A) = \mathbb{E}(\text{Var}(A|N)) + \text{Var}(\mathbb{E}(A|N)) \quad (5.1.1)$$

Now, $\text{Var}(A|N) = \sigma_\varepsilon^2 N^{-1}$, and therefore has an expectation of $\sigma_\varepsilon^2 \mathbb{E}(N^{-1})$. At the same time, the expectation of A is 0 regardless of the value of N , meaning the second term in Equation 5.1.1 is zero, leaving only $\text{Var}(A) = \sigma_\varepsilon^2 \mathbb{E}(N^{-1})$. It's worth identifying at this time the conditions under which $\mathbb{E}(N^{-1})$ is defined, but this will always be the case (so long as $N \neq 0$ a.s.).

Proof:

$$N \geq 1 \quad \text{a.s.}$$

$$\Rightarrow 0 < N^{-1} \leq 1 \quad \text{a.s.}$$

$$\Rightarrow 0 < \mathbb{E}(N^{-1}) \leq 1 \quad \text{a.s.}$$

Now, it is worth noting that A is only Gaussian if $\{n_t\}$ is almost surely constant. However, our claim is that $\{a_t\}$ is still white noise, just with variance $\sigma_\varepsilon^2 \mathbb{E}(N^{-1})$. To establish this, we must show that it is a zero-mean stationary process with ACVF $\gamma_A(h) = \mathbb{1}_{\{0\}}(h)\sigma^2$.

Firstly, we can quickly verify that A_t is zero mean with the Law of Total Expectation:

$$\mathbb{E}(A_t) = \mathbb{E}(\mathbb{E}(A_t|N_t)) = \mathbb{E}(0) = 0$$

Now we've already established $\text{Var}(A_t)$ to be independent of t , so we now verify that $\text{Cov}(A_r, A_s) = 0$ for $r \neq s$. The Law of Total Covariance states:

$$\text{Cov}(A_r, A_s) = \mathbb{E}(\text{Cov}(A_r, A_s | N_r, N_s)) + \text{Cov}(\mathbb{E}(A_r | N_r), \mathbb{E}(A_s | N_s))$$

Given that A_r and A_s are functions of independent random vectors $\boldsymbol{\varepsilon}_r \in \mathbb{R}^{n_r}$ and $\boldsymbol{\varepsilon}_s \in \mathbb{R}^{n_s}$, $\text{Cov}(A_r, A_s | N_r, N_s)$ must be zero. Having already established that both variables are zero-mean, the above equation reduces to:

$$\text{Cov}(A_r, A_s) = \mathbb{E}(0) + \text{Cov}(0, 0) = 0$$

Having shown that $\{A_t\}$ is still white noise, it now remains to estimate its variance. Though the sample sizes now vary, our estimate for σ_ε^2 hasn't changed:

$$\hat{\sigma}_\varepsilon^2 = (N - T)^{-1} \sum_{t=1}^T \sum_{i=1}^{n_t} (w_{ti} - \bar{w}_t)^2 \quad (5.1.2)$$

Without requiring the distribution of N_t , σ_a^2 can be estimated with:

$$\hat{\sigma}_a^2 = \frac{\hat{\sigma}_\varepsilon^2}{T} \sum_{t=1}^T \frac{1}{N_t} \quad (5.1.3)$$

While we're still maintaining the independence of N_t , it's worth noting that at no point in the above reasoning was such a characteristic used in establishing A_t as white noise. Before exploring the implications of such an observation, let's look at some sample distributions for N_t that show the importance of using this estimator as opposed to more naive ones.

Example: $N_t \stackrel{d}{=} M_t + 1, \quad M_t \sim \text{Poi}(\lambda)$

$N_t \geq 1$, a.s. so we know that $\mathbb{E}(N_t^{-1})$ exists. We derive it now:

$$\mathbb{E}(N^{-1}) = \sum_{n=0}^{\infty} \frac{e^{-\lambda} \lambda^n}{n!(n+1)} = \frac{1}{\lambda} \sum_{n=0}^{\infty} \frac{e^{-\lambda} \lambda^{n+1}}{n!(n+1)} = \frac{1}{\lambda} \sum_{n=0}^{\infty} \frac{e^{-\lambda} \lambda^{n+1}}{(n+1)!} = \frac{1}{\lambda} (1 - e^{-\lambda})$$

From this we obtain the following expression for the variance of a_t :

$$\text{Var}(a_t) = \frac{\sigma_{\varepsilon}^2(1 - e^{-\lambda})}{\lambda} \quad (5.1.4)$$

We can validate this through simulating i.i.d. samples of $N_t = M_t + 1$, with $M_t \stackrel{\text{i.i.d.}}{\sim} \text{Poi}(\lambda)$, then for each value of $t = 1, \dots, T$ we simulate N_t i.i.d. realizations of Gaussian noise ε_{ti} with variance σ_{ε}^2 . We set $\lambda = 2$, $\sigma_{\varepsilon}^2 = 10$, and $T = 1,000,000$.

Equation 5.1.4 yields a value of roughly $\sigma_a^2 = 4.323$ with the given parameters. Given these are simulated data, we have explicit observations of the vector $\mathbf{a}$, a luxury we won't have when fitting estimates to $X_t \mathbf{u} + \varepsilon$.

The set $\{a_t\}$ had a sample variance of 4.324. Turning to our estimates, Equation 5.1.4 produced an estimate $\hat{\sigma}_a^2 \approx 4.323$, while naively estimating σ_a^2 by dividing $\hat{\sigma}_{\varepsilon}^2$ by the mean sample size resulted in the estimate $\frac{\hat{\sigma}_{\varepsilon}^2}{\bar{n}} = 3.332$. This disparity highlights the need for the estimator defined above.

Now that we have an estimate for σ_a^2 , let's see how this change in estimator affects the rest of the method.

5.2 ARMA-Noise Models with I.I.D. Sample Size

When sample sizes are allowed to vary over time, an ARMA-Noise model changes:

$$w_{ti} = u_t + \varepsilon_{ti}, \quad \{u_t\} \sim \text{ARMA}(\phi, \theta, \sigma_{\eta}^2, T), \quad \varepsilon_{ti} \stackrel{\text{i.i.d.}}{\sim} N(0, \sigma_{\varepsilon}^2)$$

$$\bar{w}_{ti} = n_t^{-1} \sum_{i=1}^{n_t} w_{ti} = u_t + a_t, \quad a_t = n_t^{-1} \sum_{i=1}^{n_t} \varepsilon_{ti}$$

$$\{n_t\} \text{ are i.i.d.} \quad \Omega(N) \subseteq \{\mathbb{Z} > 0\}$$

The individual noise variance σ_ε^2 is estimated with 5.1.2. We review the variance of this estimate for the purposes of estimating Σ_o (Equation 4.2)

$$\text{Var}(\hat{\sigma}_\varepsilon^2) = \text{Var} \left[(N - T)^{-1} \sum_{t \in T_2} (n_t - 1) S_t^2 \right]$$

Note that we cannot treat $(N - T)$ as a constant as we could with a fixed sample size. We leverage the Law of Total Variance to decompose the variance:

$$\text{Var}(\hat{\sigma}_\varepsilon^2) = \mathbb{E}(\text{Var}(\hat{\sigma}_\varepsilon^2 | \{N_t\})) + \text{Var}(\mathbb{E}(\hat{\sigma}_\varepsilon^2 | \{N_t\})) \quad (5.2.1)$$

Focusing on the first term in Equation 5.2.1, we have:

$$\mathbb{E}(\text{Var}(\hat{\sigma}_\varepsilon^2 | \{N_t\})) = \mathbb{E} \left(\text{Var} \left[(N - T)^{-1} \sum_{t=1}^T \sum_{i=1}^{n_t} (w_{ti} - \bar{w}_{t.})^2 | \{N_t\} \right] \right)$$

Having conditioned on $\{N_t\}$ we obtain the following:

$$\mathbb{E}(\text{Var}(\hat{\sigma}_\varepsilon^2 | \{N_t\})) = \mathbb{E} \left((N - T)^{-2} \text{Var} \left[\sum_{t=1}^T \sum_{i=1}^{n_t} (w_{ti} - \bar{w}_{t.})^2 \right] \right)$$

Because of the independence within the set $\{S_t^2\}$ we get:

$$\mathbb{E}(\text{Var}(\hat{\sigma}_\varepsilon^2 | \{N_t\})) = \mathbb{E} \left((N - T)^{-2} \sum_{t=1}^T \text{Var} \left[\sum_{i=1}^{n_t} (w_{ti} - \bar{w}_{t.})^2 \right] \right)$$

Having previously derived the variance of S_t^2 we arrive at:

$$\mathbb{E}(\text{Var}(\hat{\sigma}_\varepsilon^2|\{N_t\})) = \mathbb{E}\left((N - T)^{-2} \sum_{t=1}^T 2(n_t - 1)\sigma^4\right) = 2\sigma^4\mathbb{E}(N - T)^{-1}$$

Let's now turn to the second term:

$$\text{Var}(\mathbb{E}(\hat{\sigma}_\varepsilon^2|\{N_t\})) = \text{Var}(\mathbb{E}((N - T)^{-1} \sum_{t=1}^T \sum_{i=1}^{n_t} (w_{ti} - \bar{w}_t)^2|\{N_t\}))$$

Conditioning on $\{N_t\}$, we can pass the expectation inside the first sum:

$$\text{Var}(\hat{\sigma}_\varepsilon^2|\{N_t\}) = \text{Var}\left[(N - T)^{-1} \sum_{t=1}^T \mathbb{E}\left(\sum_{i=1}^{n_t} (w_{ti} - \bar{w}_t)^2\right)\right]$$

With $(n_t - 1)S_t^2$ having expectation $(n_t - 1)\sigma_\varepsilon^2$, we arrive at:

$$\text{Var}(\mathbb{E}(\hat{\sigma}_\varepsilon^2|\{N_t\})) = \text{Var}((N - T)^{-1} \sum_t (n_t - 1)\sigma_\varepsilon^2)$$

We can then pull out the σ_ε^2 :

$$\text{Var}(\mathbb{E}(\hat{\sigma}_\varepsilon^2|\{N_t\})) = \sigma_\varepsilon^4 \text{Var}((N - T)^{-1}(N - T)) = 0$$

This is consistent with the fact that, whatever $\{N_t\}$ is, $\hat{\sigma}_\varepsilon^2$ is unbiased for σ_ε^2 , therefore having no variance. We're therefore left with:

$$\text{Var}(\hat{\sigma}_\varepsilon^2) = 2\sigma^4\mathbb{E}(N - T)^{-1}$$

With only one set of sample sizes, our only estimate for $\mathbb{E}(N - T)^{-1}$ is $(N - T)^{-1}$:

$$\widehat{\text{Var}}(\hat{\sigma}_\varepsilon^2) = 2\hat{\sigma}_\varepsilon^4(N - T)^{-1}$$

In the above expression, $\hat{\sigma}_\varepsilon^4 := (\hat{\sigma}_\varepsilon^2)^2$

$$\hat{\sigma}_a^2 = \hat{m}\hat{\sigma}_\varepsilon^2, \quad \hat{m} := \left(T^{-1} \sum_{t=1}^T n_t^{-1} \right) \hat{\sigma}_\varepsilon^2 \quad (5.2.2)$$

With $\hat{\sigma}_\varepsilon^2$, $\hat{\sigma}_a^2$ and $\widehat{\text{Var}}(\hat{\sigma}_\varepsilon^2)$ defined as above, the algorithm proposed by Wong and Miller can be once again applied, with necessary adjustments made to Σ_o and H .

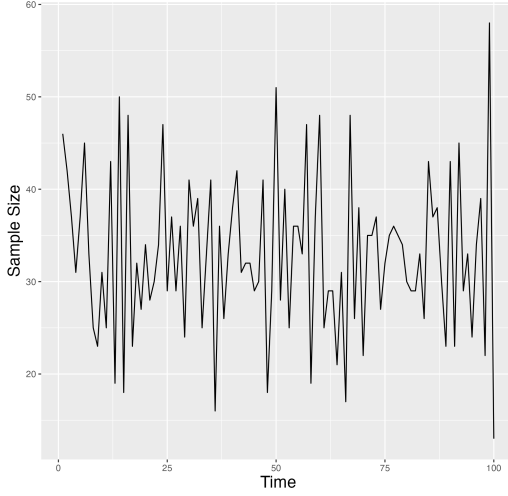
5.3 ARMA-Noise Models With Stationary Sample Sizes

Having generalized the ARIMA-Noise model proposed by Wong to allow for varying sample size, we examine the requirement that $\{N_t\}$ be independent and identically distributed, generalizing the family of distributions to which $\{N_t\}$ can belong.

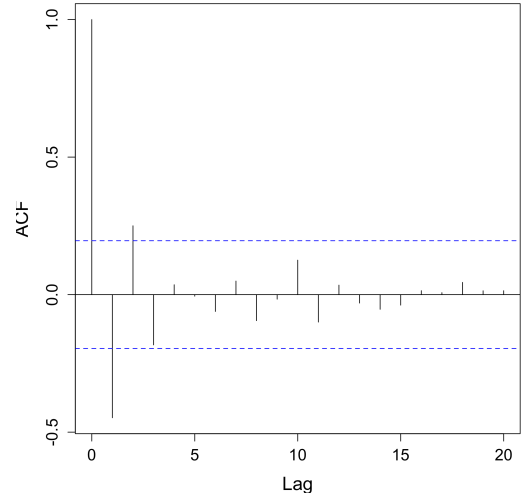
Just as it's unlikely that sample size would be exactly the same across all time steps, so too is it unlikely that daily sample sizes would be uncorrelated. It is plausible that researchers looking to maintain consistency in sample size would survey higher numbers in the event that recent previous sample sizes had been low, resulting in a negative autocorrelation. At the same time, factors affecting the ease with which data is collected might imbue the sample sizes with a positive autocorrelation.

While the introduction of temporal correlation in $\{N_t\}$ may have ramifications for $\mathbb{E}(N_t^{-1})$, the same arguments for $\{A_t\}$ being white noise still apply. An example of a natural number valued time series with autocorrelation is the Integer Valued Auto-regressive (INAR) model with binomial thinning, visualized in Figure 5.1 and defined by the following equation:

$$N_t = \alpha \circ B_{t-1} - \beta \circ B_{t-1} + P_t, \quad \alpha \circ B_t \stackrel{d}{=} \text{Bin}(N_{t-1}, \alpha), \quad P_t \sim \text{Poi}(\lambda) \quad (5.3.1)$$



(a) Sample INAR process



(b) Corresponding Sample ACF

Figure 5.1: A realization of an INAR(2) process with parameters $\alpha = 0.3$, $\beta = 0.8$, $\lambda = 50$

Such a distribution is strictly stationary and has a constant reciprocal expectation. Were we to loosen the condition of strict stationarity to include *weakly* stationary processes, it cannot be guaranteed that $\{A_t\}$ is white noise. To illustrate this, let's define the random variables X_{odd} and X_{even} :

Thus defined, X_{odd} and X_{even} have the same first and second moments, $\mathbb{E}(X_{\text{odd}}) = \mathbb{E}(X_{\text{even}}) = 2$ and $\mathbb{E}(X_{\text{odd}}^2) = \mathbb{E}(X_{\text{even}}^2) = 5$. The time series $\{N_t\}$ defined as equal in distribution to X_{odd} for odd values of t and X_{even} for even ones is therefore weakly stationary. However, X_{odd} and X_{even} have different $(-1)^{\text{st}}$ moments:

$$\mathbb{E}(X_{\text{odd}}^{-1}) = \frac{2}{3} \quad \mathbb{E}(\text{even}^{-1}) = \frac{11}{18}$$

With N_t defined as above, the daily average A_t , while independent, has dif-

fering variance at odd and even time steps, precluding it from being white noise. It therefore becomes clear that it is not so much independence in sample sizes so much as strict stationarity that ensures $\{A_t\}$ is white noise.

We therefore can extend the procedure outlined by [7] to allow for varying sample size, so long as $\{N_t\}$ is conceivably a realization of a strictly stationary integer-valued time series. $\hat{\sigma}_a^2$ is calculated using Equation 5.1.3, and its asymptotic variance estimated with 5.1.3.

5.4 ARMA-Noise Processes with Non-stationary Sample Sizes

Having now established that strictly positive sample sizes being weakly stationary does not guarantee that resulting aggregates are white noise, prompting us to seek any further generalization that would allow for deterministic changes in N_t , something that would seemingly preclude resulting aggregates from being white noise. For instance, the set of sample sizes determined by $N_t = \mathbb{1}_{2\mathbb{N}}(t)n_1 + \mathbb{1}_{2\mathbb{N}-1}(t)n_2$ would immediately result in non-stationary noise. However, so far we have only established that $\mathbb{E}(N_t^{-1})$ needs to be constant across $\{t\}$. Let's then look at a distribution with constant $(-1)^{\text{st}}$ moment and a deterministic, non-constant mean with a trend:

$$\Omega(N_t) = \{1, t\}$$

We want to construct the distribution of N_t so that $\mathbb{E}(N_t^{-1}) = c \forall t$

$$\mathbb{E}(N_t^{-1}) = P(N_t = 1) + P(N_t = t)t^{-1} = c$$

Because the cardinality of $\Omega(N_t)$ is two, this expression simplifies to:

$$\mathbb{E}(N_t^{-1}) = (1 - P(N_t = t)) + P(N_t = t)t^{-1} = c$$

Defining p_t to be $P(N_t = t)$, we find:

$$p_t = \frac{t(c-1)}{1-t}$$

Since $t \geq 1$ for all t , c must be less than or equal to one. We therefore flip the direction of subtraction in the numerator and denominator:

$$p_t = \frac{t(1-c)}{t-1}$$

We can quickly double check that $\mathbb{E}(N_t^{-1})$ is in fact constant:

$$\mathbb{E}(N_t^{-1}) = 1 - p_t + \frac{p_t}{t} = 1 - \frac{t(1-c)}{t-1} + \frac{1-c}{t-1}$$

$$\mathbb{E}(N_t^{-1}) = \frac{t-1-t(1-c)+1-c}{t-1}$$

$$\mathbb{E}(N_t^{-1}) = \frac{tc-c}{t-1} = \frac{(t-1)c}{t-1} = c$$

Having now defined the distribution of N_t , let's find the expression for its mean.

$$\mathbb{E}(N_t) = 1 - p_t + p_t t = 1 + p_t(t-1) = 1 + \frac{t(1-c)}{t-1}(t-1) = 1 + t(1-c)$$

Since $c \in (0, 1)$, so too is $(1-c)$ and we get a positive linear trend for $\mathbb{E}(N_t)$.

Having a mean that is linear in time, we clearly do not have a stationary process

in our sequence of sample sizes. However, the aggregates of the resulting noise are uncorrelated zero-mean random variables with constant variance, and as a result are still white noise. This is visualized in Figure 5.2

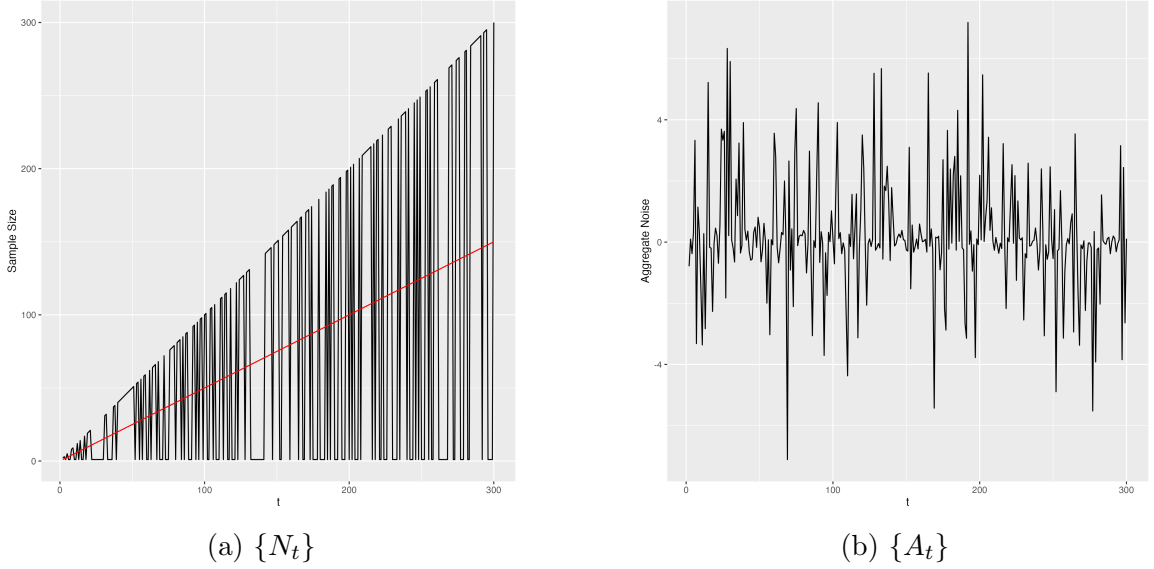


Figure 5.2: Sample Sizes $\{N_t = (t)M_{t+1} + 1\}$, where $M_t \stackrel{\text{i.i.d.}}{\sim} \text{Ber}(\frac{t}{2(t-1)})$ and resulting white noise $\{A_t = N_t^{-1} \sum_{i=1}^{N_t} \varepsilon_{ti}\}$

The takeaway from all this is that a constant expected inverse of $\{N_t\}$ is the critical piece for $\{A_t\}$ to still be white noise.

We distill this observation into the following lemma:

Let $\{\varepsilon_{ti}\}$ be i.i.d. random variables with

$$\mathbb{E}(\varepsilon_{ti}) = 0, \quad \text{Var}(\varepsilon_{ti}) = \sigma_\varepsilon^2 < \infty,$$

independent across both t and i . Let $\{N_t\}$ be a stochastic process such that

$$P(N_t \in \mathbb{N} \setminus \{0\}) = 1 \quad \text{for all } t,$$

and assume that $\mathbb{E}(N_t^{-1})$ is constant for all t .

Define

$$A_t := \frac{1}{N_t} \sum_{i=1}^{N_t} \varepsilon_{ti}.$$

Then $\{A_t\}$ is white noise with $\text{Var}(A_t) = \sigma_\varepsilon^2 \mathbb{E}(N_t^{-1})$

So long as the set of inverse sample sizes $\{n_t^{-1}\}$ is conceivably of constant expectation, the ARMA-Noise method is applicable.

Chapter 6

Computational Methods

6.1 Computational Methods for Fixed Daily Sample Size

A pitfall of treating the data in its original form, as opposed to calculating aggregates at each time step, is the relative strain this can put on computation and memory. We can leverage certain aspects of the model to make choices in our computational methods that greatly improve computing time and reduce memory requirements.

6.1.1 Inverting Ω

The biggest computational bottleneck in this procedure is the calculation and inversion of Ω , a necessary step in estimating parameters and calculating the log-likelihood. We can expedite these processes with expressions for Ω that relate its spectral decomposition to that of Γ .

We begin by recalling the expression for Ω and defining P and Λ as the components of the spectral decomposition of $\mathbf{V} = \sigma_\eta^2 \Gamma$:

$$\Omega = X_t \Gamma X_t' + \sigma_\varepsilon^2 I_N, \quad \Gamma = \sigma_\eta^2 P \Lambda P'$$

Because Γ is positive definite, all eigenvalues $\{\lambda_i\}_{i=1}^T$ are real-valued and strictly positive. Let $\{\mathbf{p}_i\}_{i=1}^T$ be the corresponding set of eigenvectors. If daily sample sizes are a constant n , then $X_t \mathbf{p}_i$ is an eigenvector of $X_t \Gamma X_t'$ with corresponding eigenvalue $n\lambda_i$:

$$(X_t \Gamma X_t')(X_t \mathbf{p}_i) = (X_t \Gamma)(X_t' X_t) \mathbf{p}_i = X_t \Gamma n I_T \mathbf{p}_i = n X_t \Gamma \mathbf{p}_i = n \lambda (X_t \mathbf{p}_i)$$

Given that $\mathbf{p}_i$ is a normal vector and $X_t \mathbf{p}_i$ has n copies of each element of $\mathbf{p}_i$, then it necessarily follows that $\|X_t \mathbf{p}_i\|^2 = n$. We can then obtain a normal eigenvector with $n^{-1/2} X_t \mathbf{p}_i$ and verify that the resulting set $\{n^{-1/2} X_t \mathbf{p}_i\}$ is orthonormal:

$$\langle n^{-1/2} X_t \mathbf{p}_i, n^{-1/2} X_t \mathbf{p}_j \rangle = n^{-1/2} X_t \mathbf{p}_i v_j' X_t' n^{-1/2} = n^{-1/2} X_t \mathbf{0} X_t' n^{-1/2} = \mathbf{0}, \quad i \neq j$$

Now, $X_t \Gamma X_t'$ is an $N \times N$ matrix expressible as a $T \times T$ block matrix, for which the i, j block is $\gamma_{\mathbf{u}}(i - j) J_n$, with J_n defined as an $n \times n$ matrix of ones. As a result, $N - T$ rows of $X_t \Gamma X_t'$ are simply copies of the other T rows, meaning $\{n^{1/2} \lambda_i\}_{i=1}^T$ are the only nonzero eigenvalues of $X_t \Gamma X_t'$. Because $X_t \Gamma X_t'$ is symmetric, any vector in its null space is orthogonal to all eigenvectors corresponding to nonzero eigenvalues.

This means we just need a $N \times (N - T)$ matrix whose columns form an orthonormal basis for the null space of $X_t \Gamma X_t'$. Before defining said matrix, let's call it Q and motivate its derivation by detailing its use in cutting down computing time:

$$X_t \Gamma X_t' = \begin{bmatrix} n^{-1/2} X_t P & Q \end{bmatrix} \begin{bmatrix} n^{1/2} \sigma_\eta^2 \Lambda & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{bmatrix} \begin{bmatrix} n^{-1/2} P' X_t' \\ Q' \end{bmatrix}$$

Because the columns of $n^{-1/2} X_t P$ and Q are all orthonormal and Γ is expressible as $\sigma_\eta^2 V$, we can now give the following spectral decomposition for Ω :

$$\Omega = \begin{bmatrix} n^{-1/2} X_t P & Q \end{bmatrix} \begin{bmatrix} n^{1/2} \sigma_\eta^2 \Lambda + \sigma_\varepsilon^2 I_T & \mathbf{0} \\ \mathbf{0} & \sigma_\varepsilon^2 I_N \end{bmatrix} \begin{bmatrix} n^{-1/2} P' X_t' \\ Q' \end{bmatrix} \quad (6.1.1)$$

We now derive an expression for the inverse of Ω , for which we only have to invert a $T \times T$ diagonal matrix:

$$\Omega^{-1} = \begin{bmatrix} n^{-1/2} X_t P & Q \end{bmatrix} \begin{bmatrix} (n^{1/2} \sigma_\eta^2 \Lambda + \sigma_\varepsilon^2 I_T)^{-1} & \mathbf{0} \\ \mathbf{0} & \sigma_\varepsilon^{-2} I_N \end{bmatrix} \begin{bmatrix} n^{-1/2} P' X_t' \\ Q' \end{bmatrix}$$

$$\Omega^{-1} = n^{-1} X_t P (n^{1/2} \sigma_\eta^2 \Lambda + \sigma_\varepsilon^2 I_T)^{-1} P' X_t' + \sigma_\varepsilon^{-2} Q Q'$$

Having motivated its derivation, we now define Q . Let v be a vector in the null space of $X_t \Gamma X_t'$. By definition v must be orthogonal to every row vector in $X_t \Gamma X_t'$, and due to symmetry every column vector. If orthogonal to every column vector, v must be orthogonal to any vectors that form a basis for the column space, namely the eigenvectors forming the columns of $n^{-1/2} X_t P$. If we're able to find $N - T$ vectors that are orthonormal to each other and the eigenvectors previously defined, we will necessarily have constructed a basis for the null space of $X_t \Gamma X_t'$, and in doing so obtained the missing $N - T$ eigenvectors thereof. Let us first note that the matrix $X_t P$ can be expressed as the following block matrix:

$$X_t P = \begin{bmatrix} \vdots \\ P_t \\ \vdots \end{bmatrix}, \quad P_t \in \mathbb{R}^{T \times T} \quad (P_t)_{i,j} = (\mathbf{p}_j)_t$$

Noting that the entries of P_t are independent of i , the column space of P_t is equal to that of $\mathbf{1}_n$. If we define $\{\mathbf{q}_i\}_{i=1}^{n-1}$ to be an orthonormal basis for $\mathcal{N}(\mathbf{1}_n)$ and $\mathbf{Q}_t$ to be the $n \times (n-1)$ matrix with column vectors $\{\mathbf{q}_i\}$, we then define the matrix $\mathbf{Q}$ as follows:

$$\mathbf{Q} = \begin{bmatrix} \ddots & & \mathbf{0} \\ & \mathbf{Q}_j & \\ \mathbf{0} & & \ddots \end{bmatrix}, \quad j = 1, \dots, T$$

The columns of $\mathbf{Q}_j$ are necessarily orthogonal to those of $\mathbf{Q}_i$ for $i \neq j$ since no pair of corresponding entries are both nonzero, and the columns of $\mathbf{Q}_j$ are orthogonal to each other since the columns of $\mathbf{Q}_{j,t}$ are orthogonal and all other entries are zero. It is for this same reason that the columns of $\mathbf{Q}_i$ are of magnitude 1. We can now express $\mathbf{Q}\mathbf{Q}'$ as the computationally cheap block diagonal matrix:

$$\mathbf{Q}\mathbf{Q}' = \begin{bmatrix} \ddots & & \mathbf{0} \\ & \mathbf{Q}_j\mathbf{Q}_j' & \\ \mathbf{0} & & \ddots \end{bmatrix}$$

Equipped with this expression for Ω^{-1} , we demonstrate how it expedites the various parts of the estimation algorithm.

6.1.2 Estimating $\mathbf{u}$

Though it is no longer used for parameter estimation, it is still valuable to estimate $\mathbf{u}$ for the purposes of identifying any short term trends that would otherwise

be lost if only fitting a linear trend. Recall $\hat{\mathbf{u}}$ is given by $\mathbf{\Gamma}X_t'\Omega^{-1}(\mathbf{y} - X_f\hat{\beta})$, which is now expressible as:

$$\hat{\mathbf{u}} = \sigma_\eta^2(P\Lambda P'X_t')(n^{-1}X_tP(n\sigma_\eta^2\Lambda + \sigma_\varepsilon^2I_T)^{-1}P'X_t' + \sigma_\varepsilon^{-2}QQ')(\mathbf{y} - X_f\hat{\beta})$$

$$\hat{\mathbf{u}} = n^{-1}\sigma_\eta^2P\Lambda(n\sigma_\eta^2\Lambda + \sigma_\varepsilon^2I_T)^{-1}P'X_t'(\mathbf{y} - X_f\beta) + \sigma_\varepsilon^{-2}\sigma_\eta^2(P\Lambda P'X_t')QQ'(\mathbf{y} - X_f\hat{\beta})$$

By virtue of the orthogonality of the column vectors of X_tP and those of Q , the above expression simplifies to:

$$\hat{\mathbf{u}} = \sigma_\eta^2P\Lambda(n\sigma_\eta^2\Lambda + \sigma_\varepsilon^2I_T)^{-1}P'(\bar{\mathbf{y}} - n^{-1}X_t'X_f\beta)$$

The punchline of this expression is that what was originally an expression involving the calculation of the inverse of a $nT \times nT$ matrix and multiplication thereof with matrices of dimension $nT \times T$ has now been re-expressed as one involving matrices of no greater dimension than $T \times T$. With this expression we can reduce the calculation of $\hat{\mathbf{u}}$ from $O(N^3)$ to $O(T^3)$, which differs by a factor of n^3 .

6.1.3 Calculating $\hat{\beta}_{\text{OLS}}$

Just as the calculation and storage of Ω and its inverse for the purposes of estimating $\mathbf{u}$ can strain memory, so too is it a problem in the calculation of $\hat{\beta}_{\text{GLS}}$, defined as:

$$\hat{\beta}_{\text{GLS}} = (X_f'\Omega^{-1}X_f)^{-1}X_f'\Omega^{-1}\mathbf{y}$$

We can change the order of operations to minimize computing time:

$$X_f' \Omega^{-1} X_f = X_f' (n^{-1} X_t P (n^{1/2} \sigma_\eta^2 \Lambda + \sigma_\varepsilon^2 I_T)^{-1} P' X_t' + \sigma_\varepsilon^{-2} Q Q') X_f \quad (6.1.2)$$

We now distribute X_f' and X_f on the right hand side.

$$X_f' \Omega^{-1} X_f = n^{-1} X_f' X_t P (n^{1/2} \sigma_\eta^2 \Lambda + \sigma_\varepsilon^2 I_T)^{-1} P' X_t' X_f + \sigma_\varepsilon^{-2} X_f' Q Q' X_f \quad (6.1.3)$$

If we then prioritize calculating $X_f' X_t$ and $X_f' Q$, which are respectively matrices of dimension $k \times T$ and $k \times (N - T)$, we then greatly reduce the maximum size of matrix in the expression. Inverting $X_f' \Omega^{-1} X_f$.

6.2 Reducing Computing Time for Varying Daily Sample Sizes

We now adapt the above section to accommodate varying $\{n_t\}$. First we define the $T \times T$ matrix N so that $X_t' X_t = N$, with $N = \text{Diag}(n_1, \dots, n_T)$.

Next let $\mathbf{r}$ to be an eigenvector of $N^{1/2} \Gamma N^{1/2}$ with corresponding eigenvector λ . If $\mathbf{p}$ is then defined as $N^{-1/2} \mathbf{r}$, we can show that $X_t \mathbf{p}$ is then an eigenvector of $X_t \Gamma X_t$:

$$\begin{aligned} X_t \Gamma X_t' (X_t \mathbf{p}) &= X_t \Gamma X_t' (X_t N^{-1/2} \mathbf{r}) = X_t \Gamma N N^{-1/2} \mathbf{r} = X_t N^{-1/2} (N^{1/2} \Gamma N^{1/2}) \mathbf{r} \\ &= X_t \Gamma X_t' (X_t \mathbf{p}) = X_t N^{-1/2} \lambda \mathbf{r} = \lambda X_t \mathbf{p} \end{aligned} \quad (6.2.1)$$

We can quickly verify that the set $\{v_t\}$ is orthonormal:

$$||X_t \mathbf{p}||^2 = \mathbf{p}' X_t' X_t \mathbf{p} = \mathbf{r}' N^{-1/2} X_t' X_t N^{-1/2} \mathbf{r} = \mathbf{r}' N^{-1/2} N N^{-1/2} \mathbf{r} = \mathbf{r}' \mathbf{r} = 1,$$

and:

$$\langle X_t \mathbf{p}_i, X_t \mathbf{p}_j \rangle = \mathbf{p}_i' X_t' X_t \mathbf{p}_j = \mathbf{r}_i' N^{-1/2} X_t' X_t N^{-1/2} \mathbf{r}_j = \mathbf{r}_i' N^{-1/2} N N^{-1/2} \mathbf{r}_j = \mathbf{r}_i' \mathbf{r}_j = 0$$

With reasoning similar to the previous section, we find the remaining eigenvectors of $X_t \mathbf{\Gamma} X_t'$. With similar derivations we arrive at the generalized expressions of $\hat{\mathbf{u}}$, $\hat{\beta}_{\text{GLS}}$, and $l(\xi)$:

Our expression for $\hat{\mathbf{u}}$ becomes:

$$\hat{\mathbf{u}} = \Gamma N^{1/2} U (\Lambda + \sigma_\varepsilon^2 I_T)^{-1} U' N^{1/2} \bar{\mathbf{y}} \quad (6.2.2)$$

Each aspect of the algorithm that incorporates Ω^{-1} is adjusted accordingly.

Chapter 7

Simulations

In order to evaluate the efficacy of the methods proposed above, individual components of the REMARCS methodology and the algorithm in its entirety were tested on simulated RCS data.

7.1 ARMA-Noise Model Fitting

We begin with simulations of an ARMA(1,1)-Noise Model $\mathbf{y} = X_t \mathbf{u} + \boldsymbol{\varepsilon}$ parameterized by $\phi, \theta, \sigma_\eta^2, \sigma_\varepsilon^2$. Empirical confidence levels for each value-pair (ϕ, θ) are based on a hundred simulations with $\sigma_\eta^2 = \sigma_\varepsilon^2 = 1$, and $n_t = 2$. We first verify that the time series $\mathbf{w}$ defined by $w_t = \bar{\mathbf{y}}_t$, with MA parameter θ_b and innovation variance σ_b^2 , produces time series estimates with normalized covariance close to Σ_{ϕ, θ_b} . For each set of parameters, we calculate the true values of θ_b and σ_b^2 . We then fit $(\hat{\phi}, \hat{\theta}_b)$ with $\bar{\mathbf{w}}$ and construct confidence regions centered around those estimates, counting the rates at which these confidence regions capture the true parameter values. These empirical confidences are displayed in Figure 7.1.

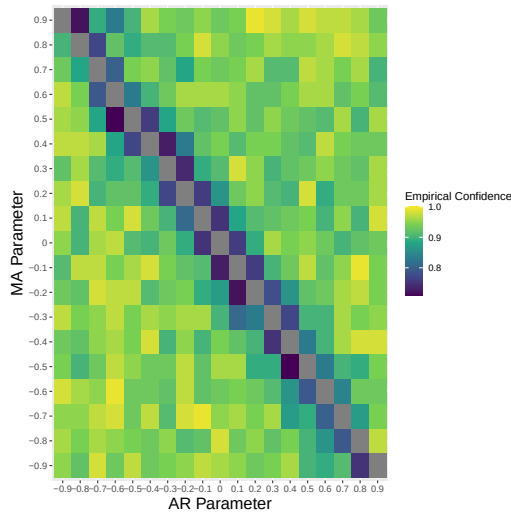
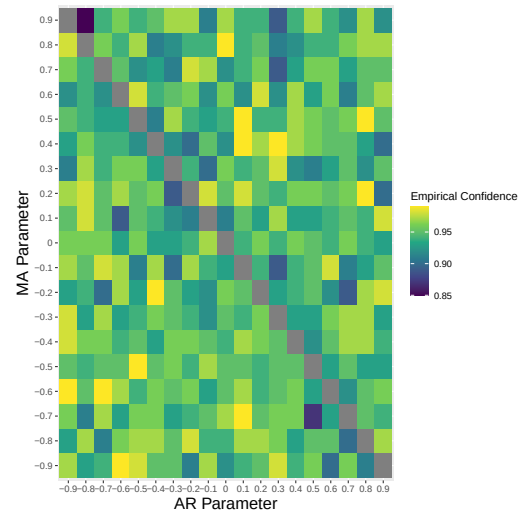
(a) $T = 1,000$ (b) $T = 10,000$

Figure 7.1: Rates at which point estimates for τ_o fall within confidence regions based on true values

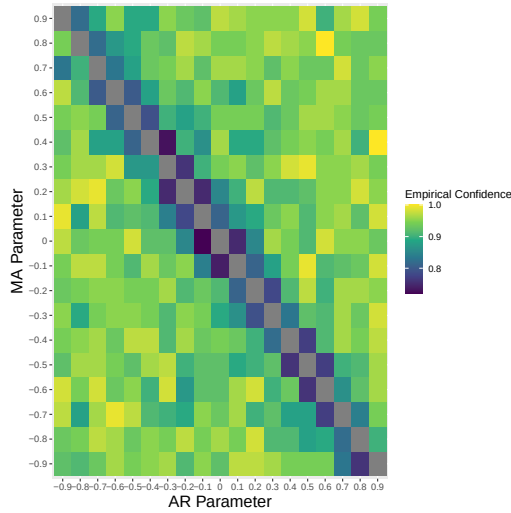
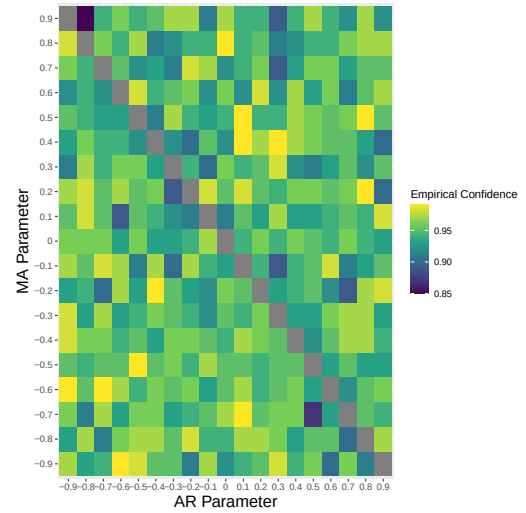
(a) $T = 1000$ (b) $T = 10,000$

Figure 7.2: Rates at which point estimates for τ_l fall within confidence regions based on sample estimates

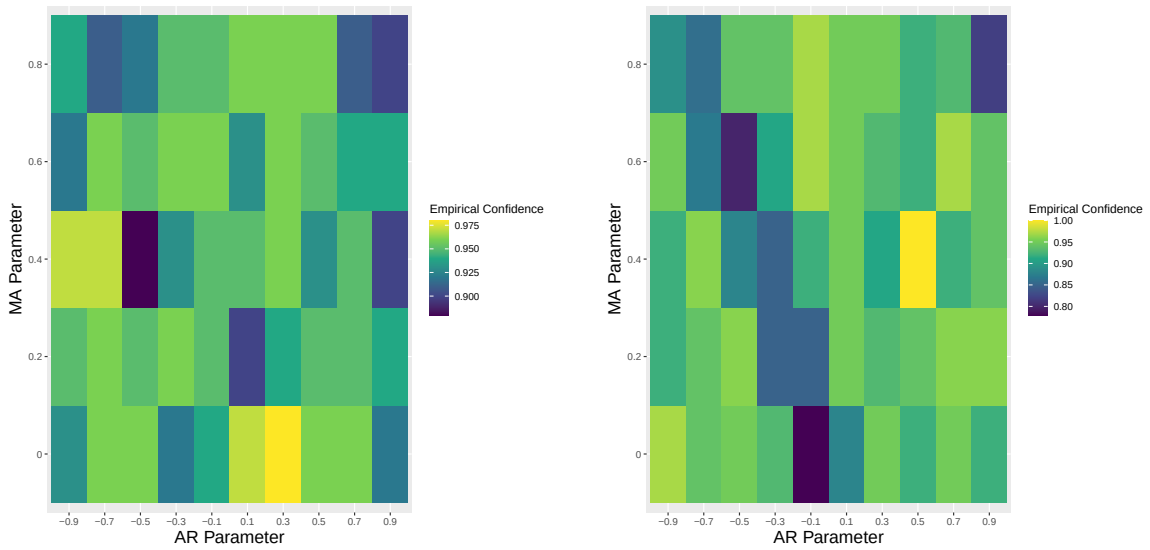
Confidence regions calculated with $\alpha = .05$ capture true values at rates close

to the predicted confidence of 95%. As we are ultimately interested in estimating the parameters $\phi, \theta, \sigma_\eta^2$, we now see how well confidence regions based on applying the Delta Method to Σ_o perform. The results in Figure 7.2 were generated by simulating realizations of an ARMA(1,1)-Noise, estimating time series parameters ϕ and θ , then using the Delta Method to construct 95% confidence intervals.

For both observable and latent parameters we can see performance dipped for parameter combinations (ϕ, θ) close to the $\phi + \theta = 0$ line for time series of length $T = 1000$. However, we can see that performance improved for $T = 10,000$, suggesting that parameter estimates for ARMA(1,1) processes that resemble white noise take longer to converge to their asymptotic distributions.

7.2 REMARCS Model Fitting

In order to assess the efficacy of this method, simulations were conducted with varying combinations of ARMA parameters (ϕ, θ) . In Figure 7.3, we can see the proportion of 100 simulations that yielded confidence regions successfully capturing $\beta = (43, -0.001, 0.25)$ (left) and $(\phi, \theta, \sigma_\eta^2)$ (right). One observation we can make is that the estimation of time series parameters appears to be hindered close to the line defined by $\phi + \theta = 0$. This under-performance is mitigated as T grows, indicating that time series more closely resembling white noise will require larger values of T for accurate estimation.



(a) Fixed Effect Parameter Estimation

(b) Time Series Parameter Estimation

Figure 7.3: Empirical confidences with which model parameters fall within confidence regions generated after model convergence

7.2.1 Development of model parameter estimates

Now that we've seen some aggregate results displaying the efficacy of this method under different parameters, we highlight some individual realizations of specific parameter combinations and observe how the parameter estimates evolve as the algorithm progresses.

Case: u and w are ARMA(1,1)

We start with a single simulation of the most general case of our model, one in which all parameters are nonzero. Exact values are given below:

T	sims	ϕ	θ	σ_η^2	σ_ε^2	β_0	β_1	β_2
1000	1	0.9	-0.6	1	10	40	-0.001	2

Table 7.1: Parameter values for simulated REMARCS model with all nonzero parameters

We begin by displaying the individual data, along with the eventual fitted values.

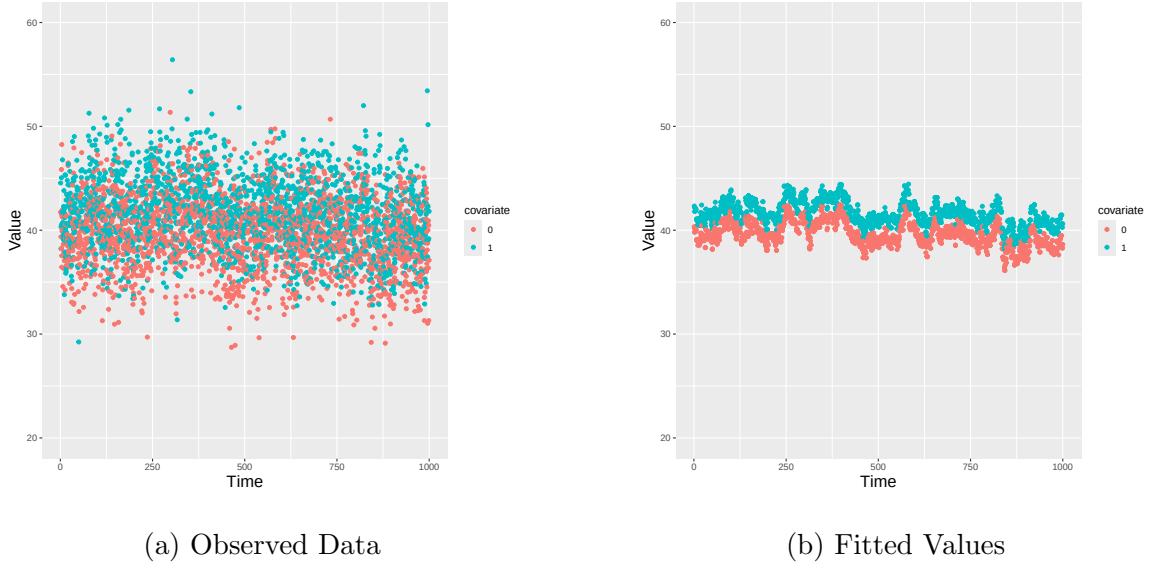
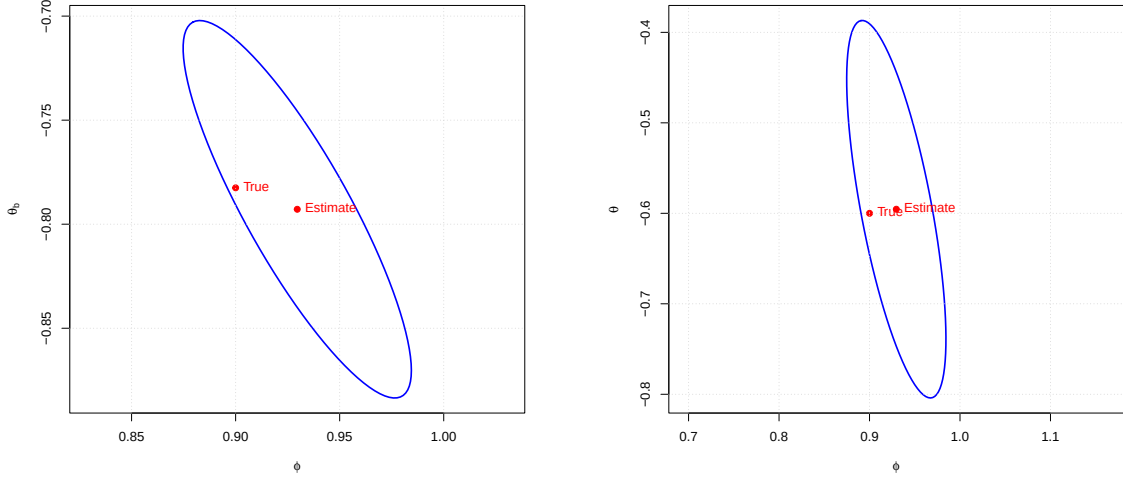


Figure 7.4: A realization of a REMARCS model before and after fitting

Once the convergence criterion of $|l(\hat{\xi}^{(k+1)}) - l(\hat{\xi}^{(k)})| < .001$ is satisfied, we take a look at the significance of “observable” time series parameter estimates $(\hat{\phi}, \hat{\theta}_b)$. Seeing that both $\hat{\phi}$ and $\hat{\theta}_b$ are significant in the left panel of Figure 7.5, we proceed to fit estimates for θ and σ_η^2 using H in Equation 3.5.9. The right panel of the same figure shows the resulting confidence region after applying the Delta Method, and we can see that the true parameters are successfully captured.



(a) Confidence region for ARMA parameters of $\mathbf{w}$

(b) Confidence region for ARMA parameters of $\mathbf{u}$

Figure 7.5: Confidence regions for time series parameter estimates of latent ARMA(1,1) process

In the table below we can see how properly accounting for temporal correlation helps update parameter estimates:

Parameter	β_0	β_1	β_2	σ_ε^2	ϕ	θ	σ_η^2
True Value	40.0000	-0.0010	2.0000	10.0000	0.9000	-0.6000	1.0000
Initial estimate	40.4306	-0.0016	1.9526	11.5826	0.0000	0.0000	0.0000
Final estimate	40.4352	-0.0017	1.9857	9.8552	0.9296	-0.5955	0.9482

Table 7.2: Parameter estimates for REMARCS model with all nonzero parameters, before and after convergence

Case: u is AR(1)

We now look at data simulated from a restricted model with the following parameters:

T	sims	ϕ	θ	σ_η^2	σ_ε^2	β_0	β_1	β_2
1000	1	0.8	0	10	10	-10	0.001	1

Table 7.3: Parameter values for simulated REMARCS model with a latent AR(1) process

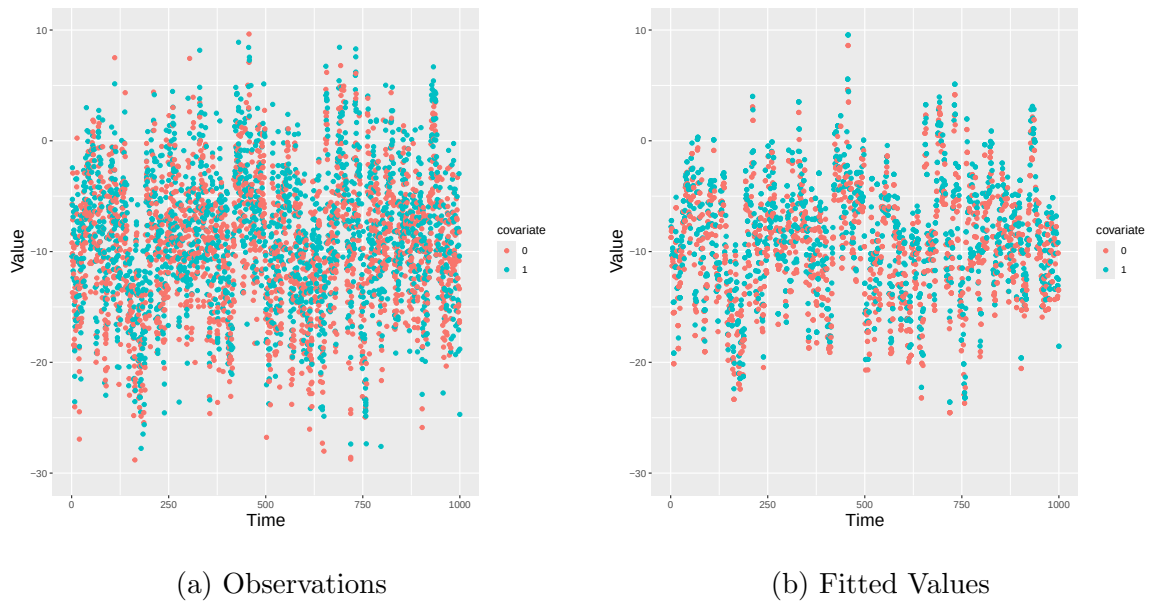
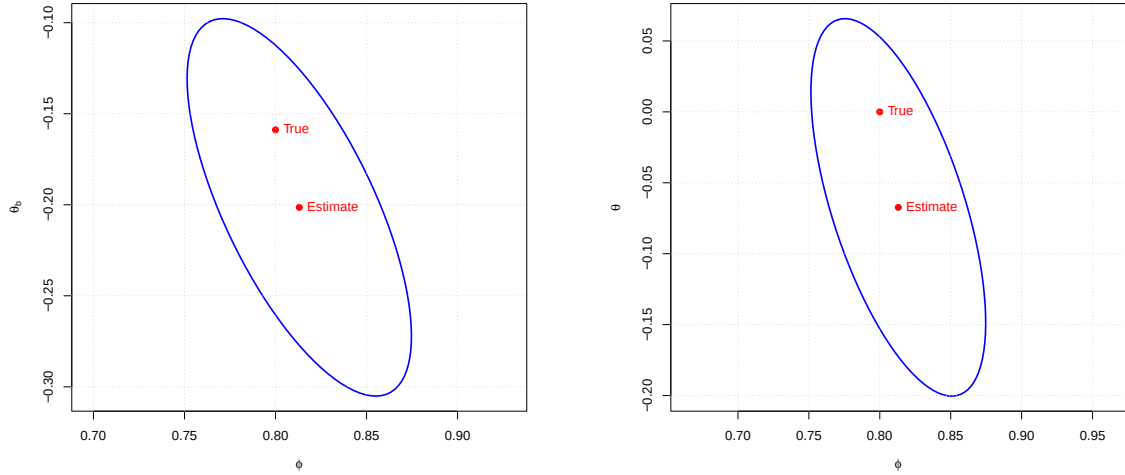


Figure 7.6: Observations and fitted values from a REMARCS model with a latent AR(1) time series

Parameter	β_0	β_1	β_2	σ_ε^2	ϕ	θ	σ_η^2
True Value	-10.0000	0.0010	1.0000	10.0000	0.8000	0.0000	10.0000
Initial estimate	-10.2365	0.0011	1.1015	38.6200	0.0000	0.0000	0.0000
Final estimate	-10.0172	0.0007	0.9519	9.8354	0.8130	-0.0673	10.7934

Table 7.4: Parameter estimates REMARCS model with latent AR(1) Time Series, before and after convergence

After having applied the general method to observations and the convergence of parameter estimates, we now perform inference on $(\hat{\phi}, \hat{\theta}_b)$. Seeing that both estimates are significant in the left side of Figure 7.6, we proceed as before with the most general form of H .



(a) Confidence Region for Parameters of $\mathbf{w}$ (b) Confidence Region for Parameters of $\mathbf{u}$

Figure 7.7: Confidence regions for time series parameter estimates of latent ARMA(1,1) process

We then see in the right panel of Figure 7.6 that the resulting estimate for θ is insignificant, prompting us to repeat the estimation of σ_η^2 by using H_θ in Equation

4.3.1. While the standard error for $\hat{\phi}$ is unaffected, the restriction of $\hat{\theta}$ will affect $\hat{\sigma}_\eta^2$. To complete this analysis, the algorithm would be restarted, now with the restriction $\hat{\theta} = 0$.

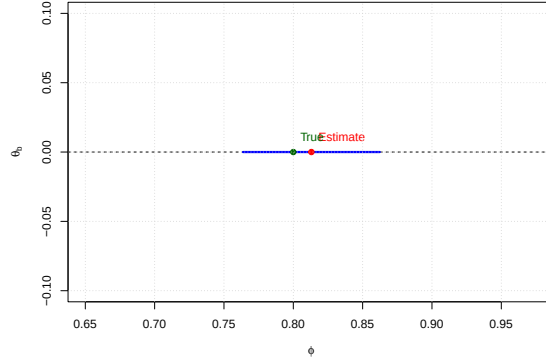


Figure 7.8: Confidence interval for ϕ after restricting model

Case: $\mathbf{u}$ is MA(1)

We now restrict $\phi = 0$ and simulate data with the following parameters:

T	sims	ϕ	θ	σ_η^2	σ_ε^2	β_0	β_1	β_2
1000	1	0	0.5	5	10	-100	0.01	3

Table 7.5: Parameter values for simulated REMARCS model with a latent MA(1) process

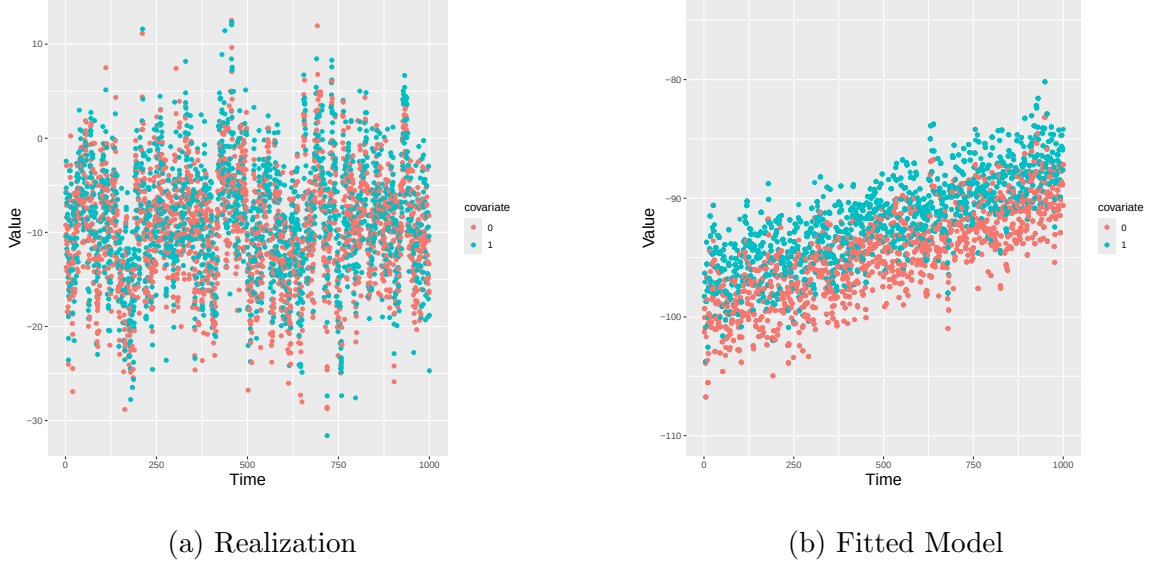


Figure 7.9: Simulated REMARCS Model Before and After Fitting

We then initiate the algorithm as usual and obtain the following parameter estimates:

Parameter	β_0	β_1	β_2	σ_ε^2	ϕ	θ	σ_η^2
True Value	-100.0000	0.0100	3.0000	10.0000	0.0000	0.5000	5.0000
Initial estimate	-100.0109	0.0103	2.9861	16.5249	0.0000	0.0000	0.0000
Final estimate	-100.0670	0.0104	2.9861	9.9510	-0.0816	0.5900	5.1878

Table 7.6: Parameter estimates for REMARCS model with latent MA(1) time series, before and after convergence

We take the final time series parameter estimates and perform inference on $\hat{\phi}$ and $\hat{\theta}_b$:

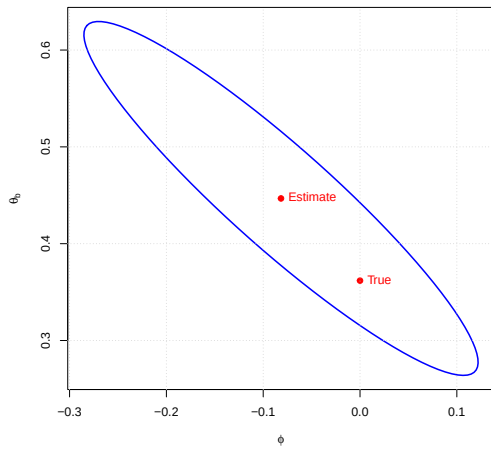
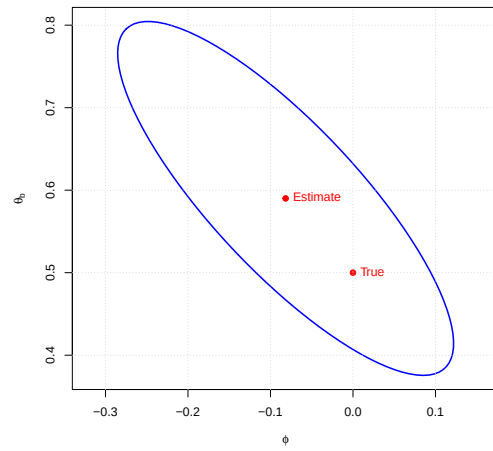
(a) Confidence region for (ϕ, θ_b) (b) Confidence region for (ϕ, θ)

Figure 7.10: Confidence regions for observable and latent moving average parameters before restricting REMARCS model with $\theta_b = 0$

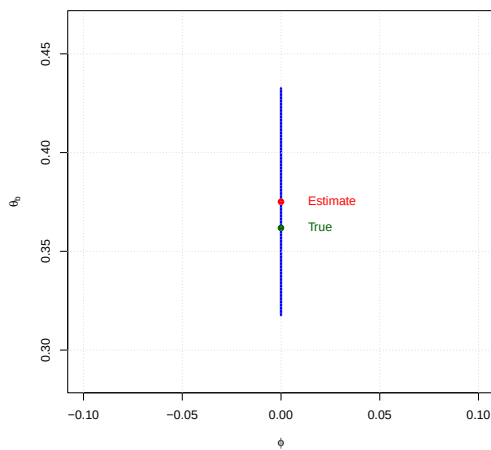
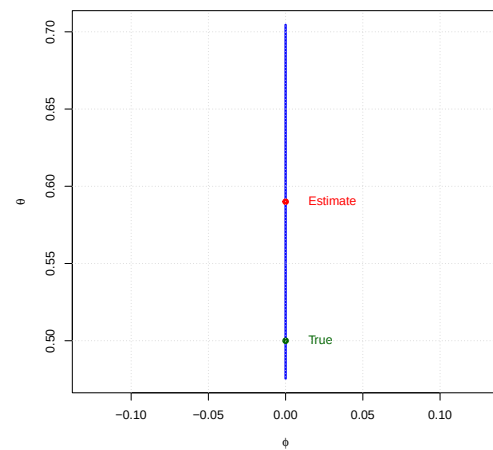
(a) Confidence region for θ_b after restricting model(b) Confidence Interval for θ after restricting model

Figure 7.11: Confidence regions for observable and latent moving average parameters after restricting REMARCS model with $\theta_b = 0$

Note that while the point estimate $\hat{\theta}$ doesn't change much, the range of the confidence interval for θ has shrunk with the decrease in degrees of freedom.

Case: w is AR(1)

We now look to the special case in which aggregate noise is an AR(1) process, which we recall can only occur if $\theta/\phi = \mathbb{E}(N^{-1})\sigma_\varepsilon^2/\sigma_\eta^2$. We select parameters for which this equality doesn't hold exactly, but is approximately true given the distribution of N_t :

T	sims	ϕ	θ	σ_η^2	σ_ε^2	β_0	β_1	β_2
1000	1	0.5	0.5	5.25	50	20	-0.001	2

Table 7.7: Parameter values for simulated REMARCS model with latent AR(1) process

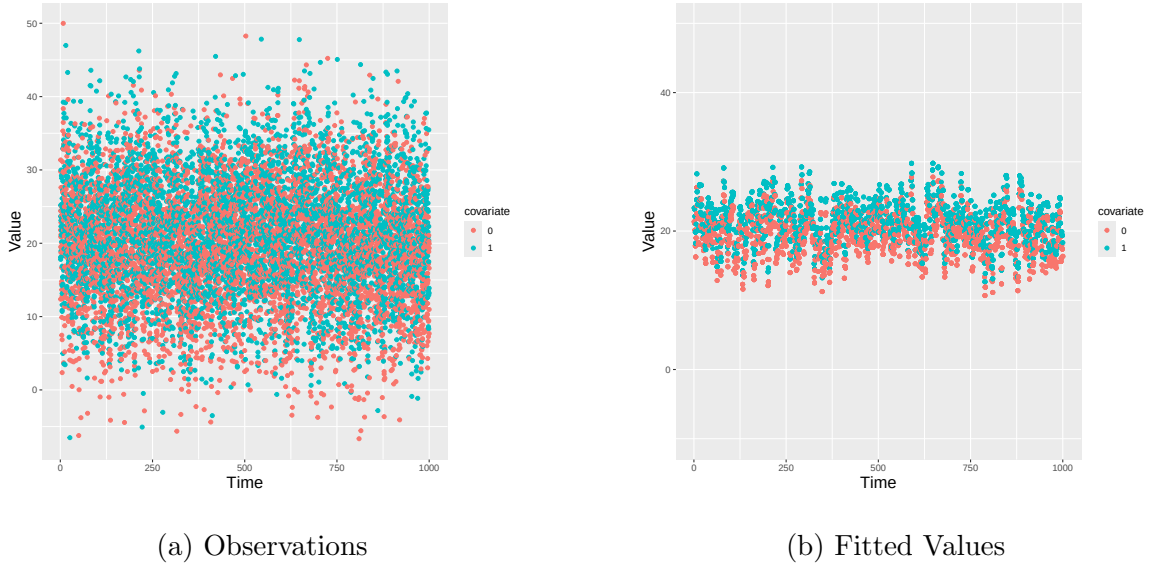


Figure 7.12: Realization of a REMARCS Model in which the sum of the latent time series and aggregated cross-sectional noise is an AR(1) process

After fitting a REMARCS model to the data we obtain parameter estimates

shown in Table 7.8 and fitted values in Figure 7.12.

Parameter	β_0	β_1	β_2	σ_ε^2	ϕ	θ	σ_η^2
True Value	20.0000	-0.0010	2.0000	50.0000	0.5000	0.5000	5.2500
Initial estimate	19.6845	-0.0007	2.0362	61.4985	0.0000	0.0000	0.0000
Final estimate	19.7719	-0.0008	1.9851	49.7772	0.4996	0.4962	5.1336

Table 7.8: Parameter estimates for REMARCS model with $\theta_b \approx 0$, before and after convergence

Note that while the point estimate for θ given by Table 7.8 is accurate, the confidence region thereof (Figure 7.13) practically spans all of $(0, 1)$, with which all that can be said is that θ is positive.

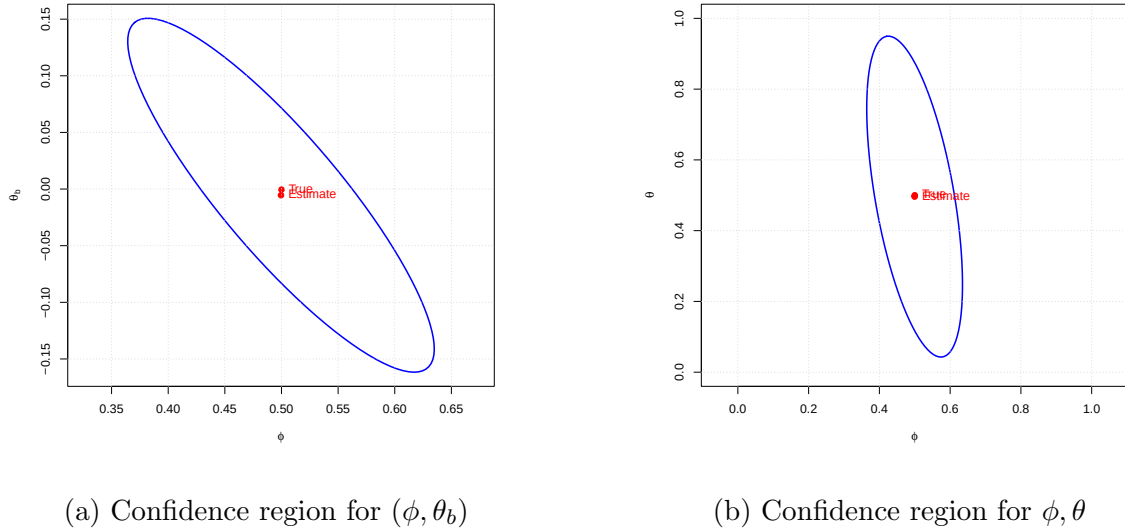


Figure 7.13: Inference on ARMA parameters of REMARCS model in which $\theta_b = 0$ with restricted model

Note in the left panel of the same figure we see that the estimate for θ_b is insignificant, prompting us to refit $\mathbf{w}$ with an AR(1) model and use H_{θ_b} (Equation

4.3.5) to construct a confidence region for $\hat{\theta}$, the products of which are shown in Figure 7.14:

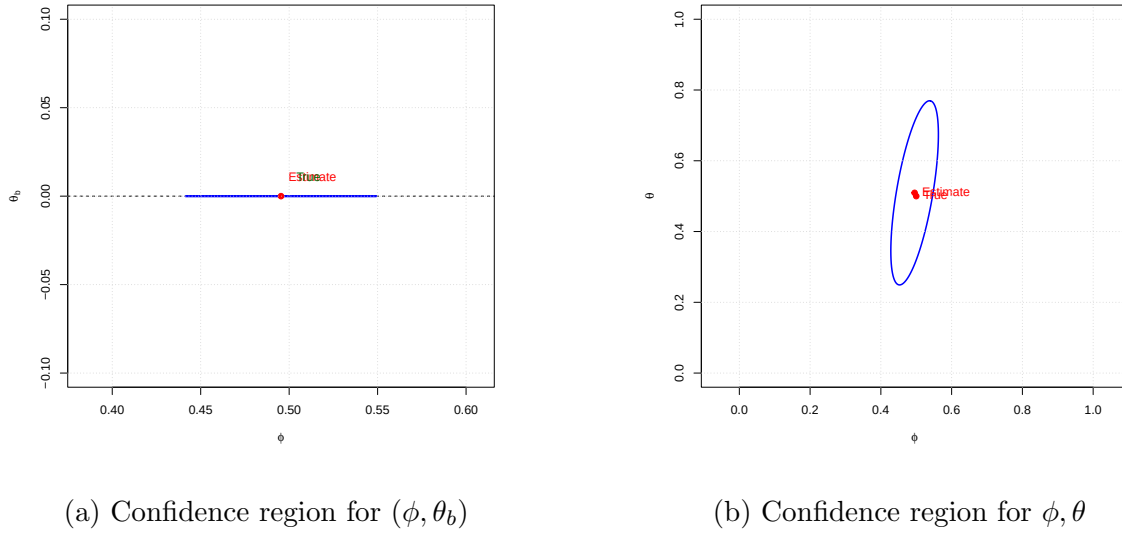


Figure 7.14: Inference on ARMA parameters of REMARCS model in which $\theta_b = 0$ with restricted model

We now obtain a confidence interval for θ markedly smaller than before. Another change of note is that the correlation between $\hat{\phi}$ and $\hat{\theta}$ has gone from negative to positive. This makes sense given that the ratio between the two now has an added restriction.

Now that our algorithm has demonstrated promising results on simulated data, we observe how it performs on real RCS survey data.

Chapter 8

WFP mVAM Data: a Case Study

Food insecurity, defined as the inability of households to gain access to sufficient, safe and nutritious food that meets dietary needs and food preferences for an active and healthy life, is a pervasive global issue. Between 2020 and 2022, the COVID-19 pandemic exacerbated global food insecurity by disrupting markets, reducing household incomes, and straining public health systems [12, 13]. Pandemic impacts were particularly severe in regions already experiencing humanitarian crises, with the disintegration of preexisting food systems resulting in increased malnutrition and susceptibility to disease [14]. An additional challenge in responding to both chronic and acute food insecurity during the COVID-19 pandemic was the inability to collect data to monitor beneficiary populations using traditional methods due to travel restrictions and quarantine orders. High-frequency phone surveys were rapidly adopted as an alternative in many countries, but the inconsistent performance of these tools highlighted the need for careful consideration in the design and implementation of remote data collection systems [5, 15]. The mobile Vulnerability Analysis and Mapping (mVAM) system developed by the World Food Programme (WFP) between 2013 and 2015 [9, 11] was one of the few remote data collection tools already in place prior to the onset of the COVID-19 pandemic, and high-frequency data from mVAM

surveys in nearly 30 countries became a critical resource for monitoring food security in populations at the regional and national level [17].

Daily participants in national phone-based surveys conducted to collect mVAM data are determined by random sampling from a large population of households, stratified at the regional level, with a different random sample taken each day to obtain a target number of responses. The standard use of high-frequency data collected by mVAM is to provide real-time tracking of household consumption levels and the use of coping strategies to mitigate food shortages, with results rapidly aggregated and smoothed using a thirty-day moving window to support WFP’s HungerMap^{LIVE} application (<https://hungermap.wfp.org/>). In addition to collecting data on consumption patterns, access to assistance, and household coping strategies, mVAM surveys also collect various demographic factors. Depending on the setting, respondents may be asked to report the size of their households, residency status, the sex of the head of household, income sources, and education. As a result, mVAM data enable not only the estimation of long-term trends, but also offer the potential to assess the impact of household-level covariates on reported food insecurity.

Food security indicators

Standard analyses of mVAM survey data focus on two key food security indicators, the Food Consumption Score (FCS) and the Reducing Coping Strategies Index (rCSI). Initially developed by WFP in 1996, the FCS is a summary statistic intended to measure dietary adequacy from both a nutritional and caloric standpoint [18]. Respondents are asked to state the number of days in the previous week that members of their household consumed foods from a standard list of groups, based on the assumption that meals in the household are communal. The FCS is computed by taking a weighted average of the days of consumption of eight major food groups (Starches, Vegetables, Fruits, Pulses, Dairy, Animal Proteins, Fats, and Sugar), according to

the following formula: $FCS = (Starches \times 2) + (Pulses \times 3) + Vegetables + Fruit + (Meat \times 4) + (Dairy \times 4) + (Fats \times .5) + (Sugar \times .5)$. The weights sum to 16, and the maximum attainable score is 112. In its reporting, WFP defines FCS thresholds to classify households into three consumption levels. In Honduras, households scoring at or below 28 are considered to have a poor level of consumption, as a score this low would be associated with consuming little other than starches, fats, and sugar on a daily basis. Households scoring between 29 and 42 have inadequate or borderline levels of consumption, while households scoring 43 or above are considered to have adequate consumption [18].

The rCSI is a food security indicator developed by the WFP and CARE to measure the frequency and severity of food-related coping strategies used by households when faced with insufficient food or resources to purchase food [20]. It is based on a seven-day recall period covering five standard behaviors: (1) relying on less preferred or less expensive foods; (2) borrowing food or relying on help from friends or relatives; (3) limiting portion sizes at mealtimes; (4) restricting consumption by adults so that children can eat; and (5) reducing the number of meals eaten per day. Each behaviour is assigned a severity weight (1, 2, 1, 3, and 1, respectively), and the household's rCSI is calculated by multiplying the reported frequency (days) of each behaviour by its weight and summing across all five strategies. Higher scores indicate greater reliance on coping strategies and, therefore, poorer food security. Although there are no universally implemented thresholds for rCSI scores, the Integrated Food Security Phase Classification (IPC) considers a score of 19 or above to represent moderate food insecurity [6].

8.1 mVAM in Honduras

The dataset considered for our case study comprises surveys collected in Honduras between April 2020 and September 2023, during which time 54,208 responses were recorded following informed consent protocols established by the World Food Programme. An average of roughly 45 households per day were randomly selected, with varying numbers of responses recorded at the department (regional) level. Demographic information for each household included geographic location at the district level (including whether the location was rural, a small city, or a large city), the gender and age of the respondent, the sex and education level of the head of household, the size of the household, and the household's primary and secondary income sources. Questions about the household's ability to access medical care and educational services were also included due to COVID-19 restrictions. Questions about food consumption and coping strategies needed to construct the FCS and rCSI indicators were included, and respondents were also asked to report whether they had received any type of assistance from the government or NGOs. These data were deidentified by WFP analysts and shared with our research team via an agreement with the US Agency for International Development in August 2023.

In addition to experiencing economic disruptions due to COVID-19, Honduras was hit by two major storms, Hurricanes Eta and Iota, in November 2020. Our analysis focuses on 5 departments that were significantly impacted by major flooding and landslides: Colón, Cortés, Gracias a Dios, Santa Bárbara, and Yoro [1]. Based on regional aggregates, nearly 20% of the households surveyed in these departments had inadequate consumption through fall 2021 while over 40% of the households engaged in crisis or emergency-level food coping during the same time period, with notable improvements in both indicators observed in 2022 (Figure 8.1). However, economic, political and environmental shocks continued to impact the country, with an uptick

in food security observed in early 2023 [19].

8.2 Modeling Results

While the summaries produced by WFP provide insights into aggregate trends, at the household level the average standard deviation in FCS scores was 15 points, indicating extremely high variability. For this reason, it is important to understand the key factors determining individual consumption levels. We began by fitting a standard regression model, with results indicating that variability in FCS scores was significantly associated with location (both department and population density), education level, household size, head of household sex, primary occupation, and receipt of assistance. We also included a time variable with both linear and squared terms to fit an overall quadratic trend.

Residuals from the fitted model, which explained 13% of the variability in the data, were approximately normally distributed, but clearly displayed a high degree of temporal correlation as shown in Figure 8.2. After running the REMARCS algorithm, model residuals were no longer correlated (see Figure 8.3). The ARMA(1,1) model estimated by REMARCS had parameters $\hat{\phi} = 0.98, \hat{\theta} = -0.85$, indicating strong autocorrelation with the potential for abrupt directional shifts. The REMARCS model also had a smaller residual standard error, decreasing from 15.51 for the OLS model to 15.07.

While the REMARCS model retained the same covariates as statistically significant as for the OLS model, there were some differences in model coefficients and their associated standard errors (see Table 8.1). In both cases, the fitted models indicated significant differences in average consumption levels across the five departments. Households in small cities had higher average consumption than rural households (set as the baseline level and included in the intercept term), which in turn had higher

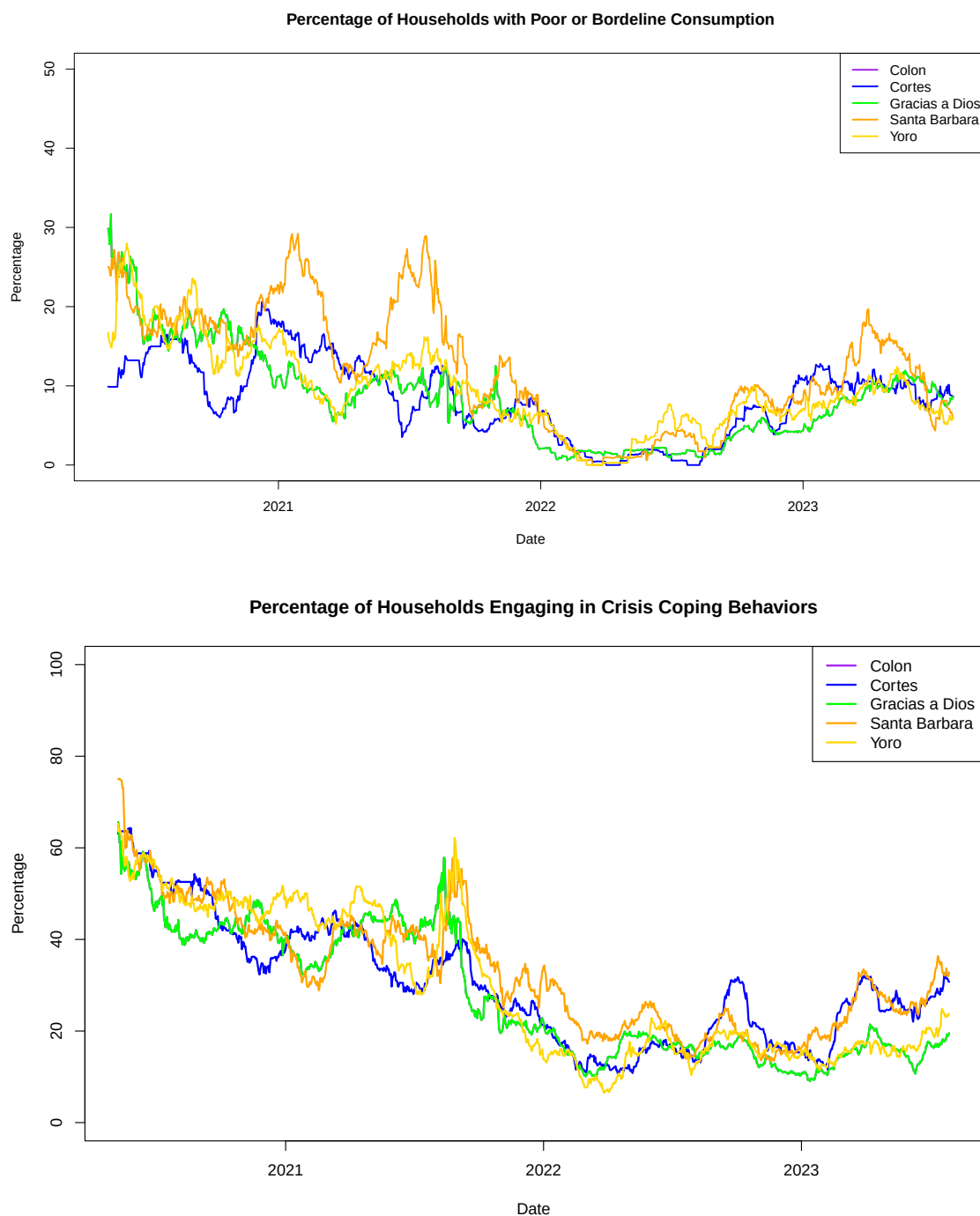


Figure 8.1: WFP estimates from aggregated FCS scores (top) and rCSI scores (bottom) in five hurricane-impacted departments

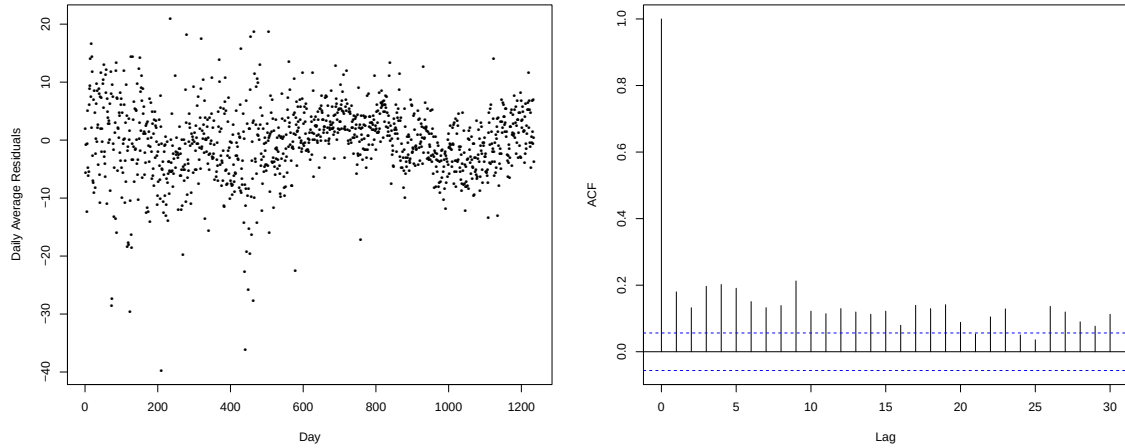


Figure 8.2: Left: Average daily residuals from the fitted regression model for FCS. Right: Autocovariance plot for average daily residuals

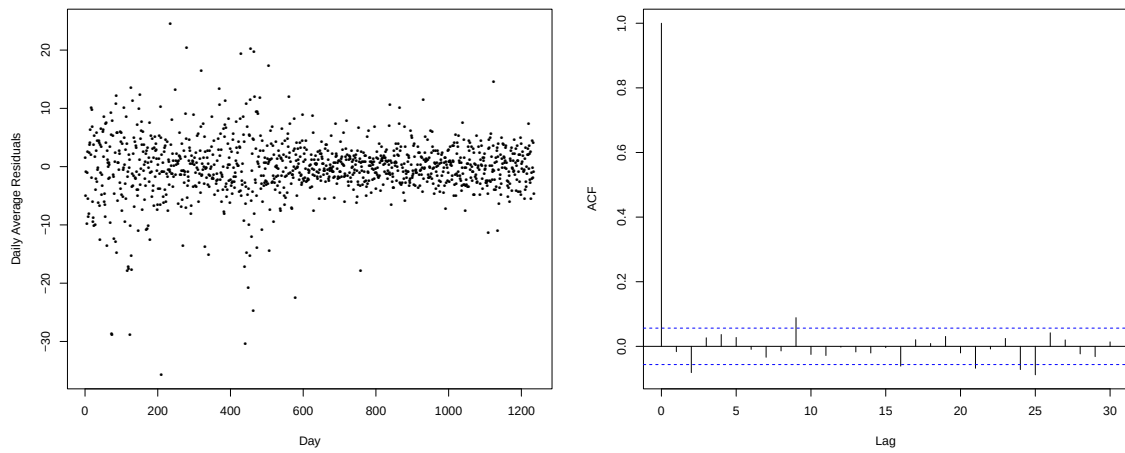


Figure 8.3: Left: Average daily residuals from the fitted REMARCS model for FCS. Right: Autocovariance plot for average daily residuals

average consumption than residents of large cities. Education was highly significant and followed an increasing trend, with the highest consumption levels estimated for households with education beyond the primary level (illiteracy was the baseline level). Households headed by both sexes fared better than either male (baseline) or female-headed households, while consumption levels were inversely associated with household size. Households receiving assistance, lacking a regular income source, or relying on charity/NGO support had lower average consumption than those with regular formal employment (informal daily labor was the baseline level). Notable changes in parameters associated with the REMARCS model were increases in the magnitudes of parameters associated with department, population density, and assistance.

To illustrate the differences between the two models, we compare predictions for a “typical” respondent in C  lon, characterized as a rural household with five individuals, headed by a male with a primary education and informal daily labor as the primary income source. The OLS model predicted an FCS score gradually increasing from 59 in spring 2020 to 63 in late 2022 before declining to 53 in 2023. The REMARCS model, on the other hand, predicted a much more volatile trajectory, with large drops in late 2020 and late 2022 that likely reflected a response to environmental and economic shocks (see Figure 8.4).

Covariate	OLS Estimate	OLS Std. Error	REMARCS Estimate	REMARCS Std. Error
Intercept	55.55892	0.69983	56.00315	1.99251
Department: Cortés	1.49768	0.38220	1.48454	0.38490
Department: Gracias a Dios	-1.29787	0.44037	-1.37484	0.44108
Department: Santa Bárbara	-2.00598	0.39614	-2.06755	0.39374
Department: Yoro	-2.04978	0.38499	-2.08285	0.38261
Day	0.01799	0.00150	0.01627	0.00714
Day^2	-0.00002	0.00000	-0.00002	0.00001
Pop Density: Large City	-1.11153	0.35924	-1.57302	0.35999
Pop Density: Small City	1.24400	0.30561	0.90564	0.30517
Education: Primary	4.63793	0.48226	4.69604	0.48044
Education: Secondary	8.03498	0.49925	7.93902	0.49882
Education: University	11.15058	0.62775	11.10039	0.62533
Education: Vocational Training	9.18409	0.65478	9.32931	0.65285
Household Size	-0.26260	0.05162	-0.26451	0.05150
Head of Household Sex: Both	0.92264	0.29633	0.95552	0.29590
Head of Household Sex: Female	-0.56173	0.35912	-0.34456	0.35684
Assistance Received	-0.88059	0.28658	-1.29160	0.28714
Primary Income: Business/SelfEmployed	4.77979	0.49561	4.82818	0.49346
Primary Income: Housecleaning/CareWork	-1.69638	0.69574	-1.94788	0.69167
Primary Income: GovAssistance/Pension	-0.29584	1.24207	-0.63630	1.23308
Primary Income: Informal Trade	0.33543	0.48890	0.43442	0.48585
Primary Income: None	-8.43743	0.56936	-8.34208	0.56743
Primary Income: Remittances/Family/Friends	2.95203	1.03774	2.61032	1.03059
Primary Income: Salaried Work	3.95280	0.33419	3.91366	0.33364
Primary Income: Charity/NGO/WFP	-13.54970	7.76009	-13.76163	7.71296
ϕ (AR parameter)	NA	NA	0.97660	0.00964
θ (MA parameter)	NA	NA	-0.85316	0.02823

Table 8.1: Caption: Parameter Estimates and Standard Errors from OLS and REMARCS Models for FCS

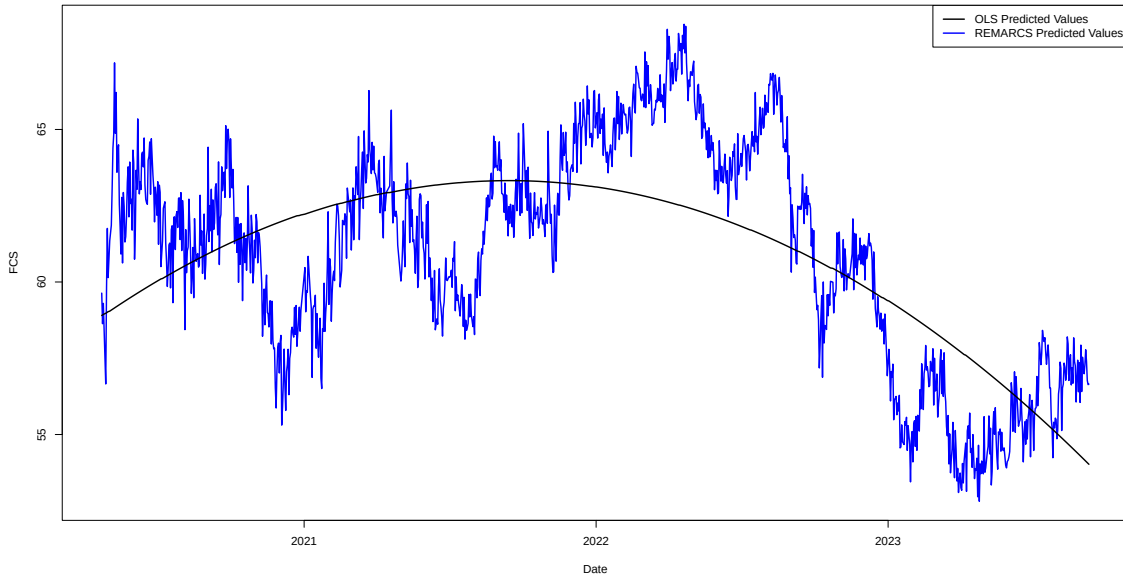


Figure 8.4: Predicted FCS values for a typical household from OLS and REMARCS models

For the rCSI, models were fit after taking the square root of the observations to correct for skewness. The OLS model explained 23% of the variability in $\sqrt{rCSI}$ scores, but once again, residuals indicated significant autocorrelation. This temporal dependence was corrected through the REMARCS model fit (Figures 8.5 and 8.6). The estimated ARMA(1,1) model was similar to that for FCS, with $\hat{\phi}=0.96$ and $\hat{\theta}=-0.83$, and the residual standard error decreased from 1.25 to 1.23.

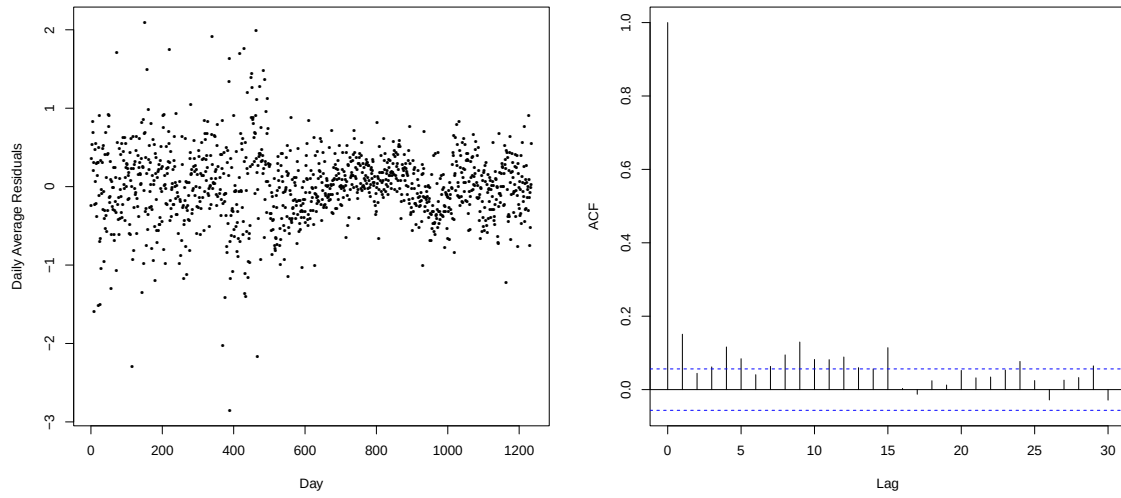


Figure 8.5: Left: Average daily residuals from the fitted regression model for $\sqrt{rCSI}$. Right: Autocovariance plot for average daily residuals

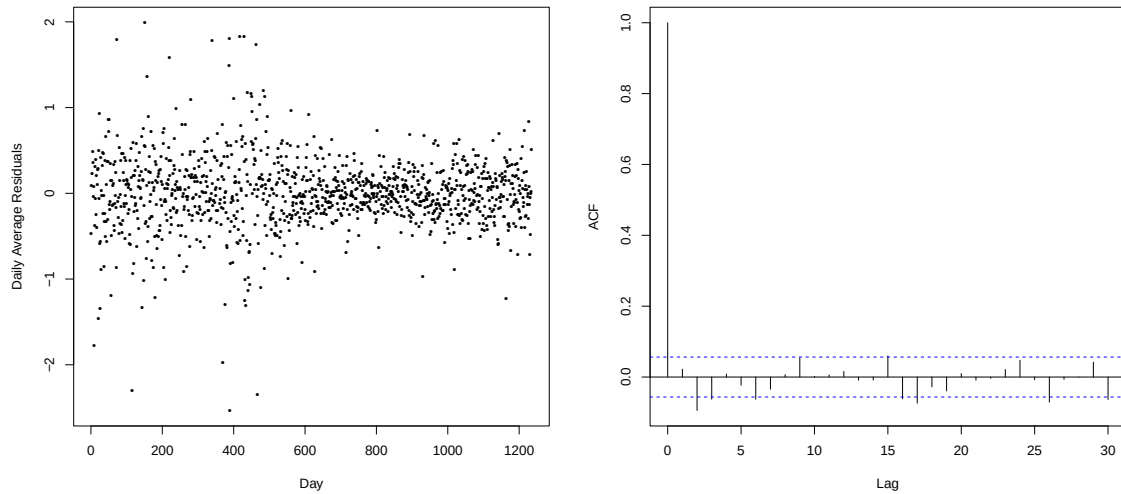


Figure 8.6: Left: Average daily residuals from the fitted REMARCS model for $\sqrt{rCSI}$. Right: Autocovariance plot for average daily residuals

Model coefficients differed in some important ways than for the FCS model, with lower average coping levels estimated for urban residents (in both small and large

cites) than for rural households (Table 8.2). Specifically, female-headed households were predicted to have significantly higher $\sqrt{rCSI}$ scores than male-headed households, although households headed by both sexes had the lowest average predicted scores. While coefficients associated with educational levels followed the same trends as for the FCS, significantly higher levels of food coping were predicted for households whose primary income source was informal trade or housekeeping/care work. Additionally, these models included an indicator for the sex of the respondent, which reflects the phenomenon that women are more likely to report higher levels of food insecurity than men [8].

Again comparing predictions for a typical rural household in Colón, we see that the REMARCS model estimated higher levels of coping in the summers of each year than the OLS model, possibly reflecting pre-harvest scarcity in these largely agricultural regions, while predicting seasonal declines in rCSI at the end of each year. Higher predicted rCSI scores were generally associated with lower FCS scores, although this was not always the case.

Overall, our analysis from mVAM data in Honduras demonstrates that, not surprisingly, daily food security surveys have a significant level of temporal autocorrelation that violates standard regression assumptions. By appropriately accounting for this dependence, we are able to not only more accurately estimate regression parameters and standard errors associated with food security, but also model short-term trends that reflect changing conditions due to environmental, political, and economic factors.

Covariate	OLS Estimate	OLS Std. Error	REMARCS Estimate	REMARCS Std. Error
Intercept	4.462830	0.056831	4.52E+00	1.03E-01
Department: Cortés	0.158862	0.030776	1.52E-01	3.11E-02
Department: Gracias a Dios	0.171668	0.035461	1.69E-01	3.56E-02
Department: Santa Bárbara	-0.008780	0.031886	-1.25E-02	3.18E-02
Department: Yoro	0.034962	0.030993	3.59E-02	3.09E-02
Day	-0.002090	0.000121	-2.20E-03	3.47E-04
Day^2	0.000001	0.000000	1.40E-06	2.77E-07
Pop Density: Large City	-0.531342	0.029015	-5.38E-01	2.92E-02
Pop Density: Small City	-0.186378	0.024603	-2.00E-01	2.47E-02
Education: Primary	-0.297135	0.038856	-2.90E-01	3.88E-02
Education: Secondary	-0.687499	0.040244	-6.90E-01	4.03E-02
Education: University	-0.957933	0.050564	-9.65E-01	5.06E-02
Education: Vocational Training	-0.995288	0.052762	-9.89E-01	5.28E-02
Household Size	0.078719	0.004156	7.72E-02	4.17E-03
Head of Household Sex: Both	-0.271312	0.023904	-2.84E-01	2.40E-02
Head of Household Sex: Female	0.176365	0.030291	1.72E-01	3.02E-02
Respondent Sex: Female	0.103970	0.021713	1.10E-01	2.17E-02
Assistance Received	0.096715	0.023070	8.76E-02	2.32E-02
Primary Income: Business/SelfEmployed	-0.604215	0.039922	-5.97E-01	3.99E-02
Primary Income: Housecleaning/CareWork	0.295960	0.056121	2.89E-01	5.60E-02
Primary Income: GovAssistance/Pension	-0.276054	0.099955	-2.80E-01	9.97E-02
Primary Income: Informal Trade	0.169133	0.039364	1.68E-01	3.93E-02
Primary Income: None	-0.133396	0.045835	-1.16E-01	4.58E-02
Primary Income: Remittances/Family/Friends	-0.118630	0.083500	-1.31E-01	8.33E-02
Primary Income: Salaried Work	-0.364533	0.026913	-3.57E-01	2.70E-02
Primary Income: Charity/NGO/WFP	0.058769	0.624449	1.81E-02	6.23E-01
ϕ (AR parameter)	NA	NA	0.955443	0.018940
θ (MA parameter)	NA	NA	-0.834486	0.039169

Table 8.2: Caption: Parameter Estimates and Standard Errors from OLS and REMARCS Models for $\sqrt{rCSI}$

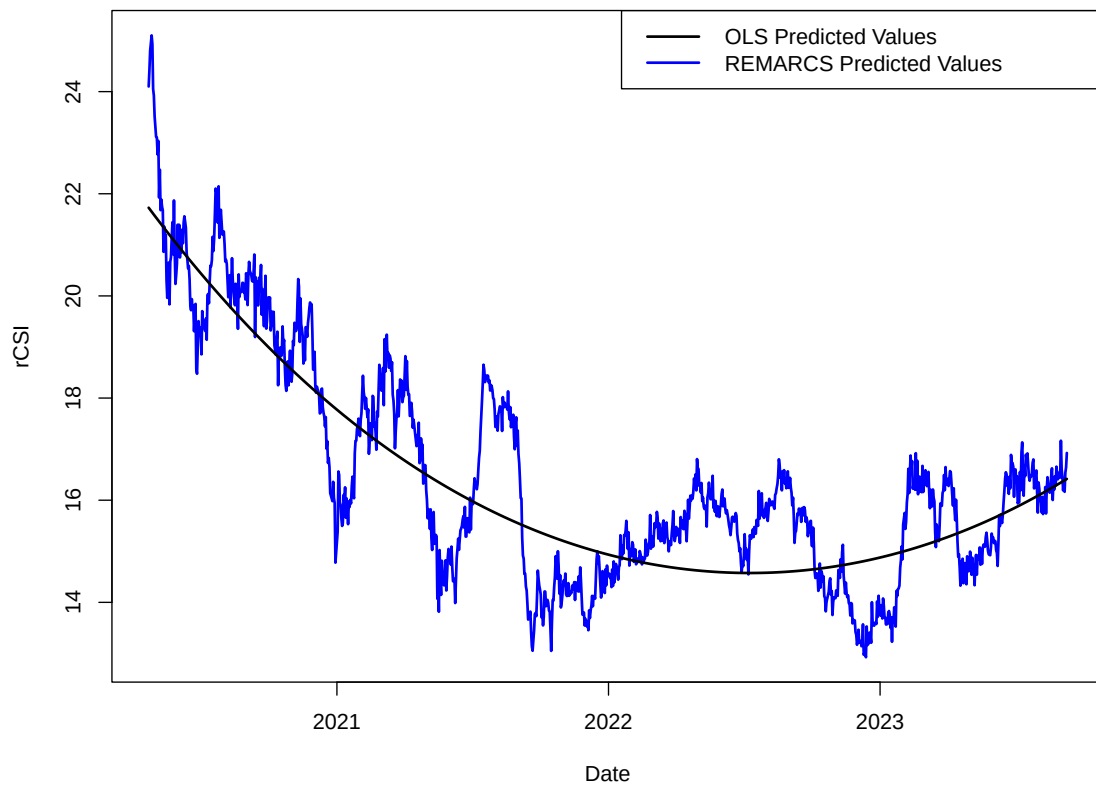


Figure 8.7: Predicted $\sqrt{rCSI}$ values for a typical household from OLS and REMARCS models

Chapter 9

Discussion

This dissertation develops a framework for analyzing high-frequency repeated cross-sectional survey data that exhibit temporal correlation while containing individual-level covariates that are time independent. It does so by introducing a latent ARMA process, inducing an overall ARMA-Noise residual structure, and recursively fitting parameter estimates with GLS and ARMA-Noise methods. In order to accommodate survey data with varying daily sample size, this research furthered ARMA-Noise modeling by relaxing the preexisting condition that sample sizes be constant. Results provide insight into how temporal correlation interacts with cross-sectional noise, particularly how not properly accounting for this interaction can obfuscate results and hinder proper inference. Given how large RCS datasets can be, computational methods were tailored to the specific characteristics of this model by leveraging the relationship between the spectral decompositions of the various covariance matrices involved. The exploration of these models has produced insight into the limitations of their analysis and yielded several interesting directions for further research.

A result central to this work is that parameter estimates for a latent ARMA process will be fundamentally flawed if fit to aggregate residuals, as the presence of additional white noise introduces bias in estimates for not only the variance parameter

but the moving average parameter as well. Such an impact is tractable, however, and can be reverse engineered to obtain asymptotically unbiased estimates for the true time series parameters. The relation between the observable and latent time series must be incorporated into the analysis of time series estimate. This impact can change the order of the ARMA process on top of the parameters themselves. For instance, in the case that the latent time series is actually an AR(1) process, the observed aggregate errors will be ARMA(1,1) with a moving average parameter of opposite sign as that of the autoregressive parameter. Modeling the noisy time series as an AR(1) process would prevent a consistent estimate for ϕ from being obtained. Simulation results reinforce this conclusion. We see that as the relative variance of the additional noise increases, estimates for ϕ obtained by fitting an AR(1) model become increasingly biased, while those fit with an ARMA(1,1) model remain accurate.

This work provides a viable method for analyzing datasets such as the mVAM surveys, elucidating individual-level effects by properly accounting for temporal shocks that would otherwise obfuscate them. Such a method avoids the information loss inherent to the aggregate methods that are currently commonplace.

Furthermore, failure to account for temporal correlation when modeling repeated cross-sectional data can lead to invalid inference. At the same time, flattening the data to be analyzed into aggregates results in significant information loss at the individual level. In particular, in the event that individual-level variables like education level or income source see little variance in their prevalence at the time-step level, fitting aggregated data using a time series with covariates could easily obscure an otherwise significant effect.

This method is not without limitations, particularly given its reliance on asymptotic distributions. As such confidence regions underperform for low values of T , particularly in boundary cases like $\phi, \theta \approx \pm 1$ and $\phi + \theta \approx 0$, as the resulting

covariance structure is close to that of white noise. The imposition of an ARMA order no greater than $(1, 1)$ is certainly limiting, as it precludes REMARCS Models with all but three types of latent ARMA processes from being properly fit.

There are a number of natural extensions that arise from this work, particularly generalizing the individual level of the model to allow for random effects. Relaxing the conditions that $p, q \leq 1$ and generalizing the latent time series to allow for any general ARMA process is also a natural question. Given that survey data so often tracks binary output variables, relaxing the Gaussian condition is of further interest. Since confidence intervals are currently based on asymptotic distributions, the derivation of small-sample corrections is warranted.

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